

The ins and outs of INNER JOIN

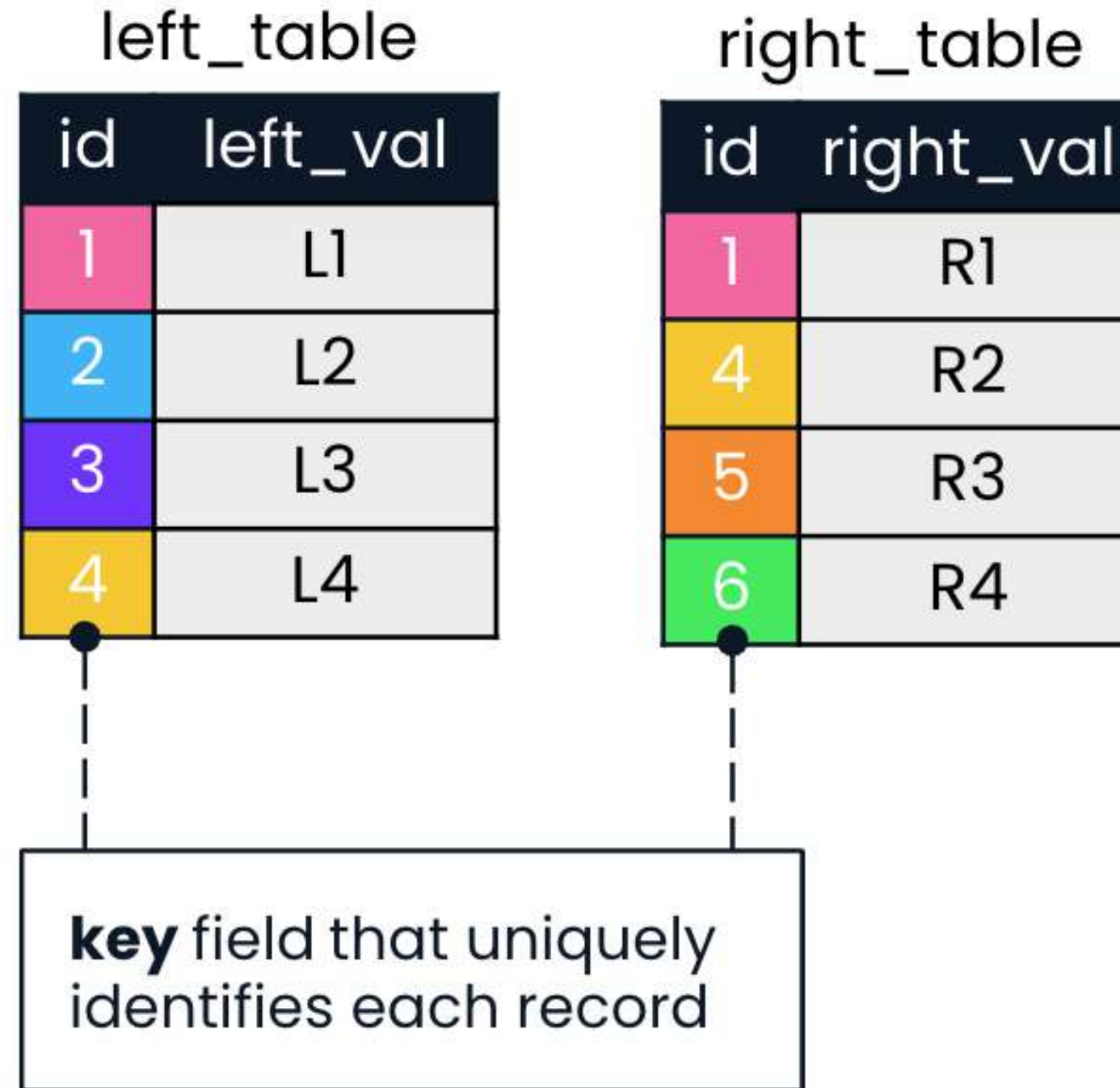
JOINING DATA IN SQL



Maham Faisal Khan

Senior Content Developer, DataCamp

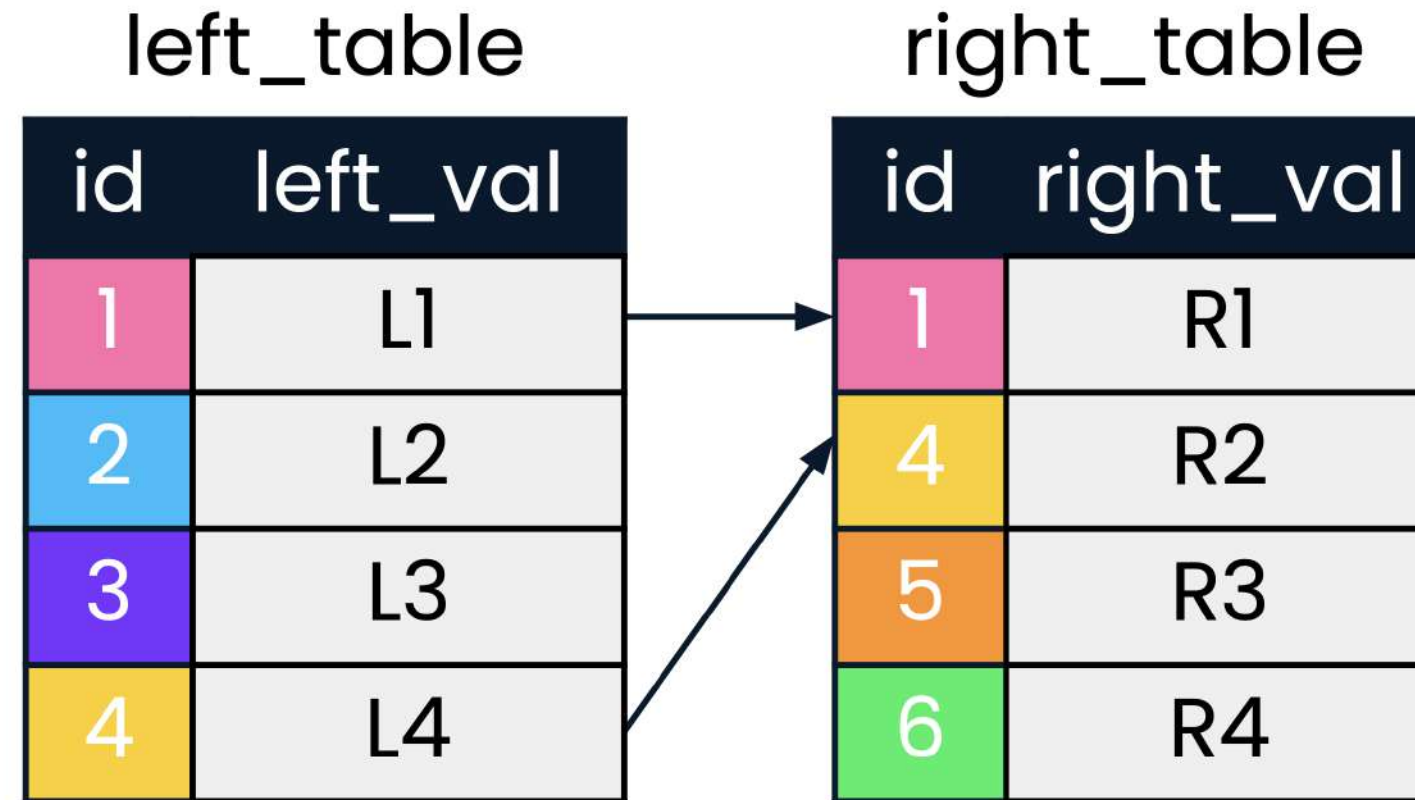
The ins and outs of INNER JOINS



The ins and outs of INNER JOINS

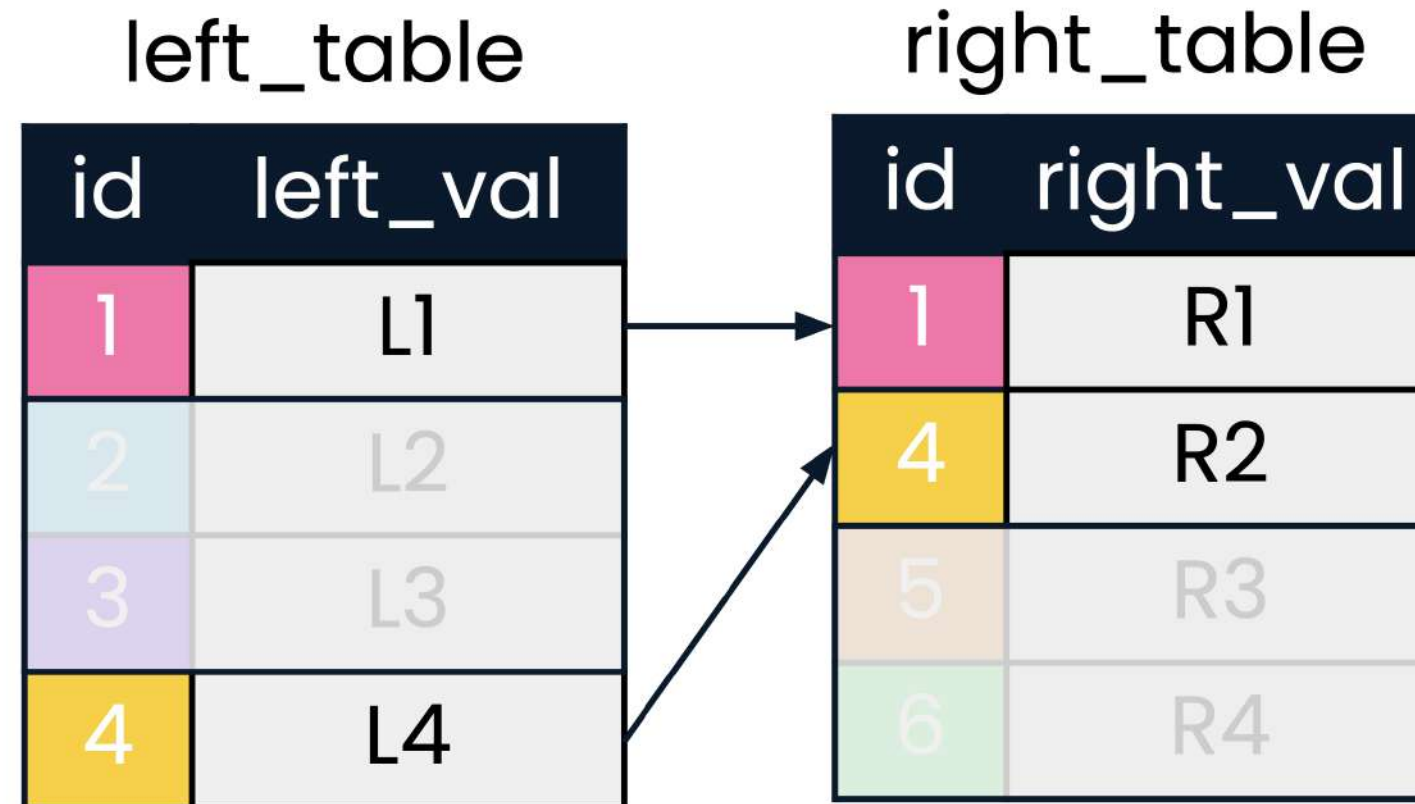
- `INNER JOIN` looks for records in both tables which match on a given field

Diagram for an `INNER JOIN` on the `id` field



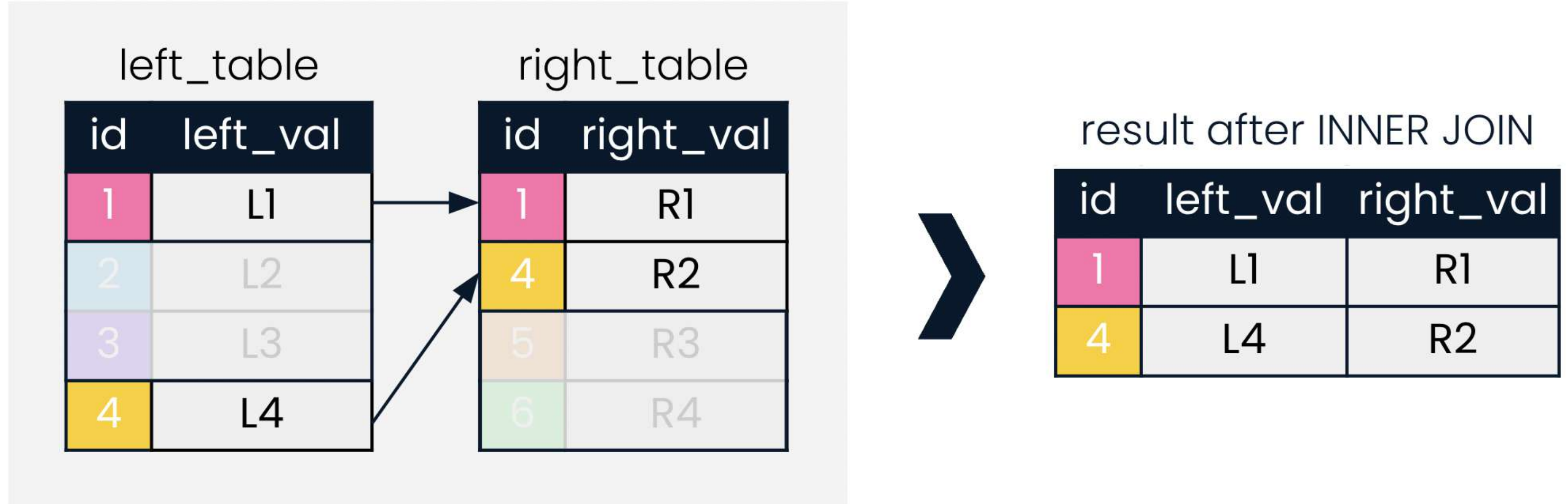
The ins and outs of INNER JOINS

Diagram for an **INNER JOIN** **ON** the **id** field



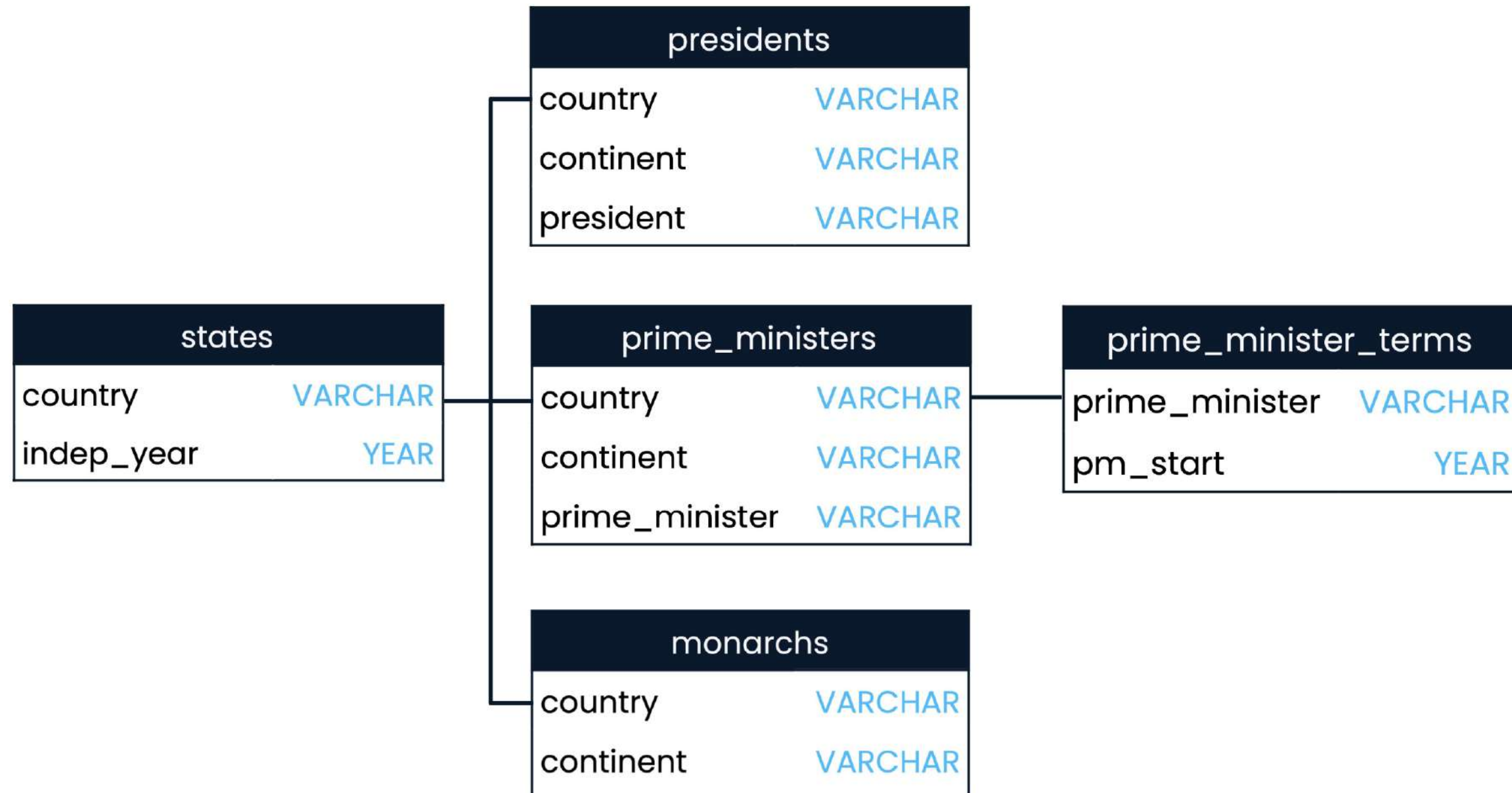
The ins and outs of INNER JOINS

Diagram for an **INNER JOIN** **ON** the **id** field



The leadership database schema

World leaders up to 2024:



At the presidents table

```
SELECT *  
FROM presidents;
```

country	continent	president
-----	-----	-----
Egypt	Africa	Abdel Fattah el-Sisi
Portugal	Europe	Marcelo Rebelo de Sousa
USA	North America	Joe Biden
Uruguay	South America	Luis Lacalle Pou
Pakistan	Asia	Asif Ali Zardari
Chile	South America	Gabriel Boric
India	Asia	Droupadi Murmu

Meet the prime ministers

The `prime_ministers` table

country	continent	prime_minister
Egypt	Africa	Mostafa Madbouly
Portugal	Europe	Luís Montenegro
Pakistan	Asia	Shehbaz Sharif
United Kingdom	Europe	Keir Starmer
India	Asia	Narendra Modi
Australia	Oceania	Anthony Albanese
Norway	Europe	Jonas Gahr Støre
Brunei	Asia	Hassanal Bolkiah
Oman	Asia	Haitham bin Tarik
New Zealand	Oceania	Christopher Luxon

Prime ministers, meet the presidents

The `presidents` table

country	continent	president
Egypt	Africa	Abdel Fattah el-Sisi
Portugal	Europe	Marcelo Rebelo de Sousa
USA	North America	Joe Biden
Uruguay	South America	Luis Lacalle Pou
Pakistan	Asia	Asif Ali Zardari
Chile	South America	Gabriel Boric
India	Asia	Droupadi Murmu

The `prime_ministers` table

country	continent	prime_minister
Egypt	Africa	Mostafa Madbouly
Portugal	Europe	Luís Montenegro
Pakistan	Asia	Shehbaz Sharif
United Kingdom	Europe	Keir Starmer
India	Asia	Narendra Modi
Australia	Oceania	Anthony Albanese
Norway	Europe	Jonas Gahr Støre
Brunei	Asia	Hassanal Bolkiah
Oman	Asia	Haitham bin Tarik
New Zealand	Oceania	Christopher Luxon

Our first INNER JOIN

```
--Inner join of presidents and prime_ministers, joining on country
SELECT prime_ministers.country, prime_ministers.continent, prime_minister, president
FROM presidents
INNER JOIN prime_ministers
ON presidents.country = prime_ministers.country;
```

Note. The `table.column_name` format must be used when selecting columns that exist in both tables to avoid a SQL error.

country	continent	prime_minister	president	
-----	-----	-----	-----	
Egypt	Africa	Mostafa Madbouly	Abdel Fattah el-Sisi	
Portugal	Europe	Luís Montenegro	Marcelo Rebelo de Sousa	
Pakistan	Asia	Shehbaz Sharif	Asif Ali Zardari	
India	Asia	Narendra Modi	Ram Nath Kovind	

Aliasing tables

```
--Inner join of presidents and prime_ministers, joining on country
SELECT p2.country, p2.continent, prime_minister, president
FROM presidents AS p1
INNER JOIN prime_ministers AS p2
ON p1.country = p2.country;
```

country	continent	prime_minister	president
Egypt	Africa	Mostafa Madbouly	Abdel Fattah el-Sisi
Portugal	Europe	Luís Montenegro	Marcelo Rebelo de Sousa
Pakistan	Asia	Shehbaz Sharif	Asif Ali Zardari
India	Asia	Narendra Modi	Ram Nath Kovind

Aliases can be used in the `table.column_name` syntax in `SELECT` and `ON` clauses.

Using USING

```
--Inner join of presidents and prime_ministers, joining on country
SELECT p2.country, p2.continent, prime_minister, president
FROM presidents AS p1
INNER JOIN prime_ministers AS p2
USING(country);
```

country	continent	prime_minister	president
Egypt	Africa	Mostafa Madbouly	Abdel Fattah el-Sisi
Portugal	Europe	Luís Montenegro	Marcelo Rebelo de Sousa
Pakistan	Asia	Shehbaz Sharif	Asif Ali Zardari
India	Asia	Narendra Modi	Ram Nath Kovind

Let's practice!

JOINING DATA IN SQL

Defining relationships

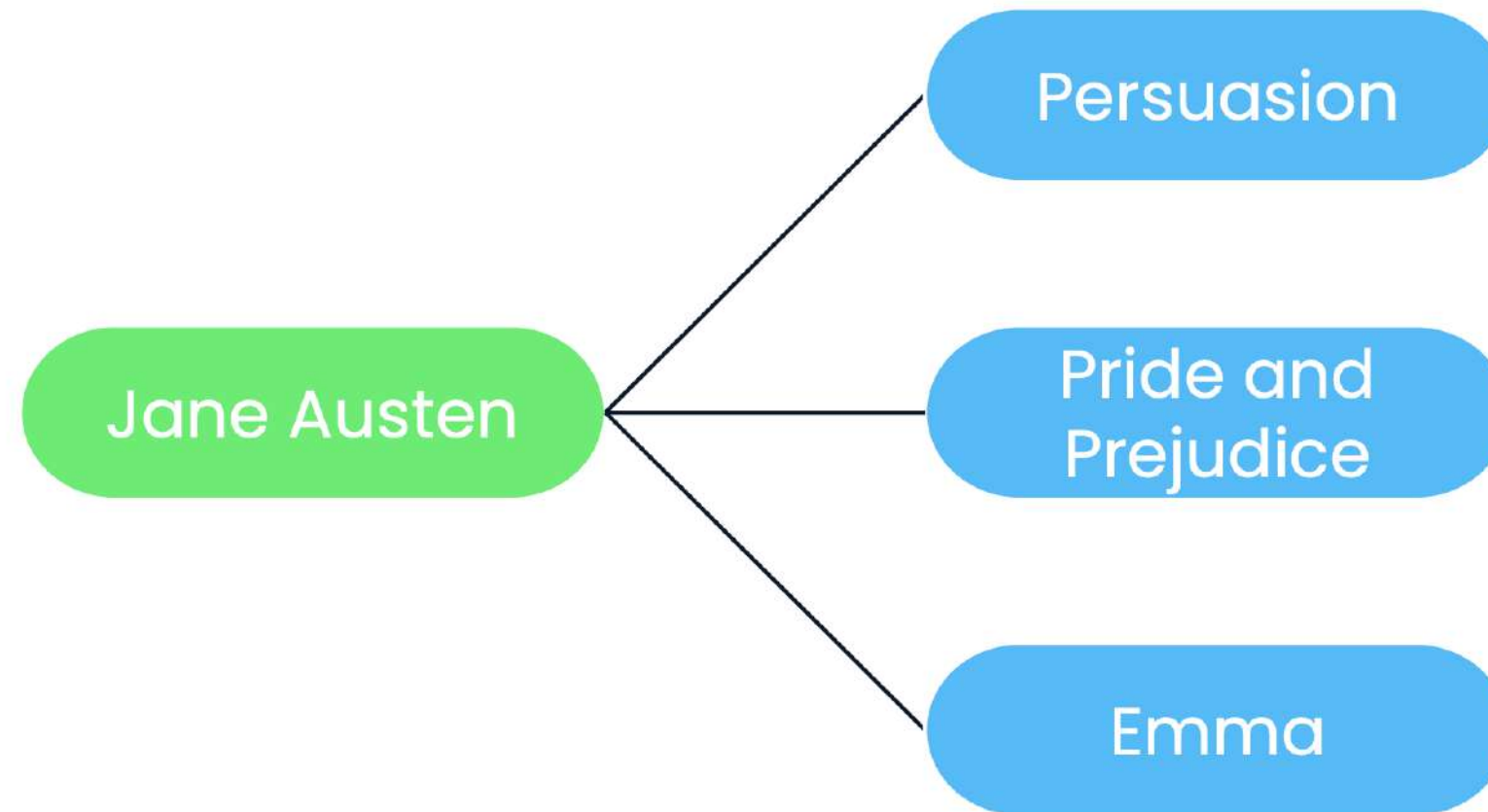
JOINING DATA IN SQL



Maham Faisal Khan

Senior Content Developer, DataCamp

One-to-many relationships



One-to-many relationships

authors	
author_id	INT
first_name	VARCHAR
last_name	VARCHAR

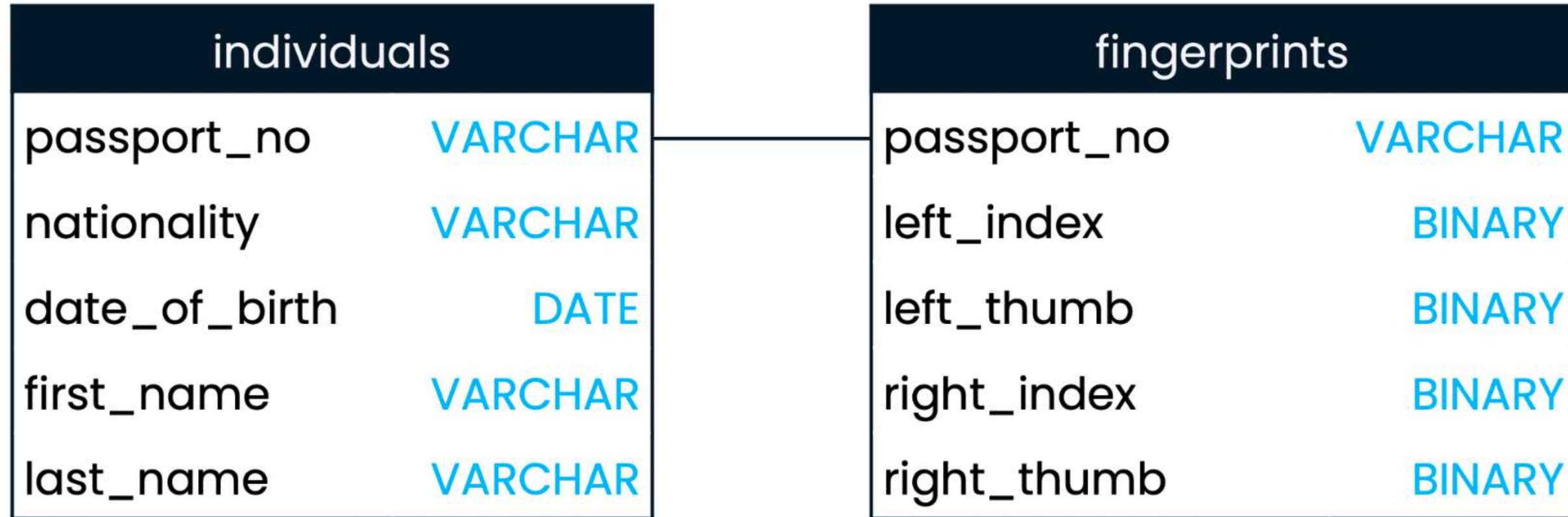


books	
book_id	INT
author_id	INT
title	VARCHAR
genre	VARCHAR
rating	FLOAT
year_published	YEAR

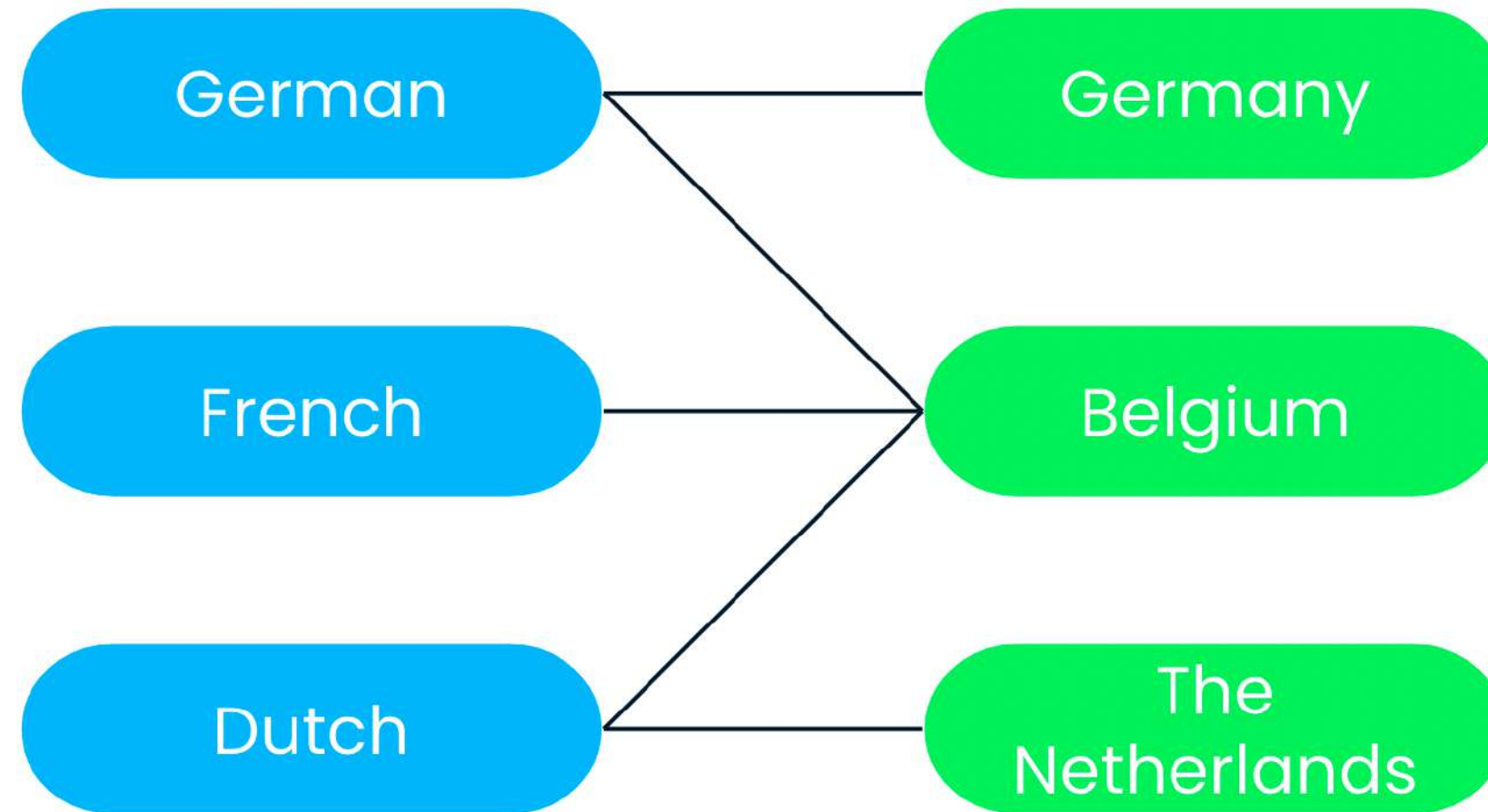
One-to-one relationships



One-to-one relationships



Many-to-many relationships



Let's practice!

JOINING DATA IN SQL

Multiple joins

JOINING DATA IN SQL



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Senior Content Developer, DataCamp

Joins on joins

```
SELECT *  
FROM left_table  
INNER JOIN right_table  
ON left_table.id = right_table.id  
INNER JOIN another_table  
ON left_table.id = another_table.id;
```

Note. Depending on the use case, `left_table` or `right_table` can be used in the `ON` clause.

Joins on joins

The `prime_minister_terms` table from our database of world leaders

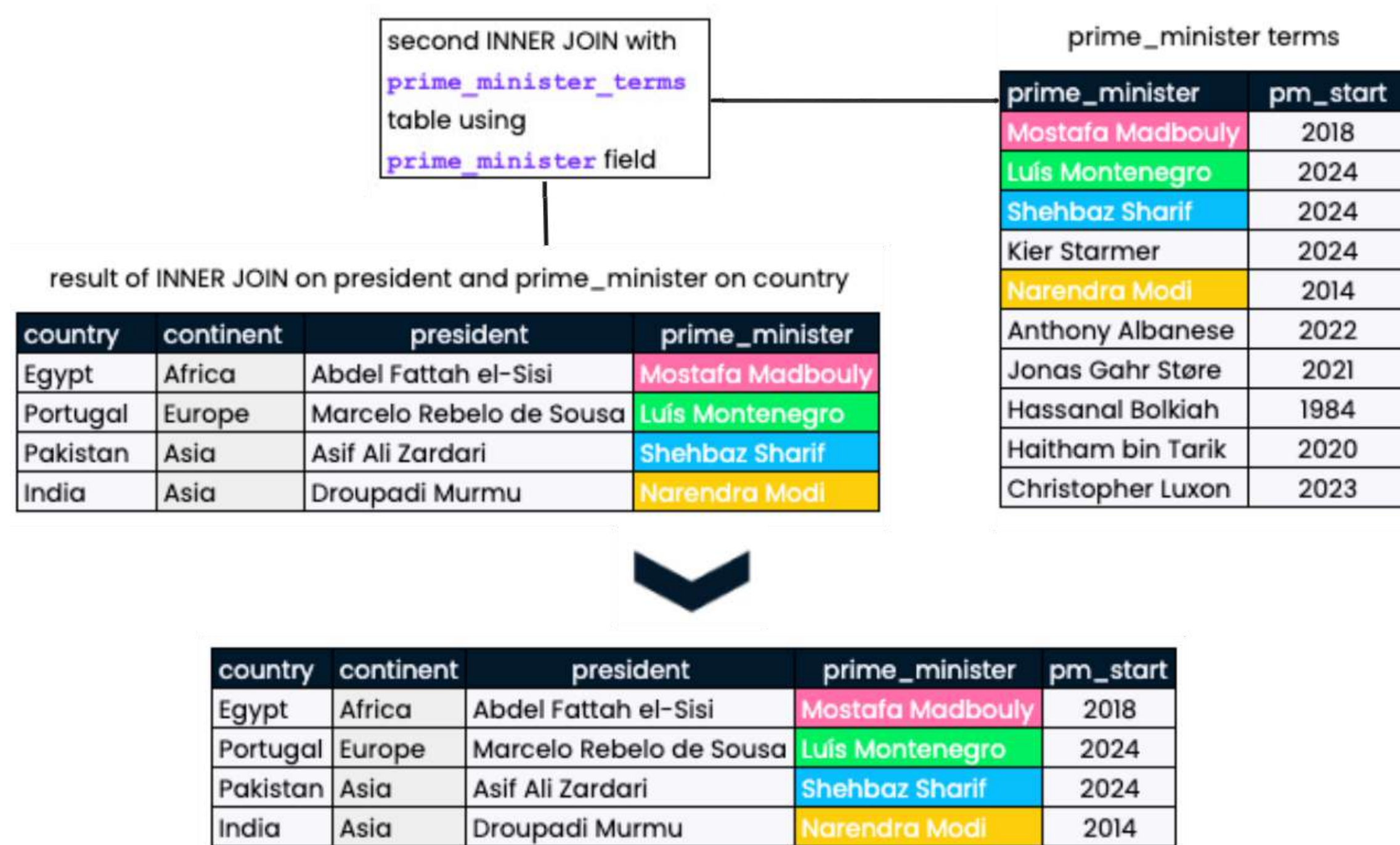
prime_minister	pm_start
Mostafa Madbouly	2018
Luís Montenegro	2024
Shehbaz Sharif	2024
Kier Starmer	2024
Narendra Modi	2014
Anthony Albanese	2022
Jonas Gahr Støre	2021
Hassanal Bolkiah	1984
Haitham bin Tarik	2020
Christopher Luxon	2023

What to join first?

```
SELECT p1.country, p1.continent,  
       president, prime_minister  
FROM prime_ministers as p1  
INNER JOIN presidents as p2  
USING(country);
```

country	continent	president	prime_minister
-----	-----	-----	-----
Egypt	Africa	Abdel Fattah el-Sisi	Mostafa Madbouly
Portugal	Europe	Marcelo Rebelo de Sousa	Luís Montenegro
Pakistan	Asia	Asif Ali Zardari	Shehbaz Sharif
India	Asia	Droupadi Murmu	Narendra Modi

Chaining joins



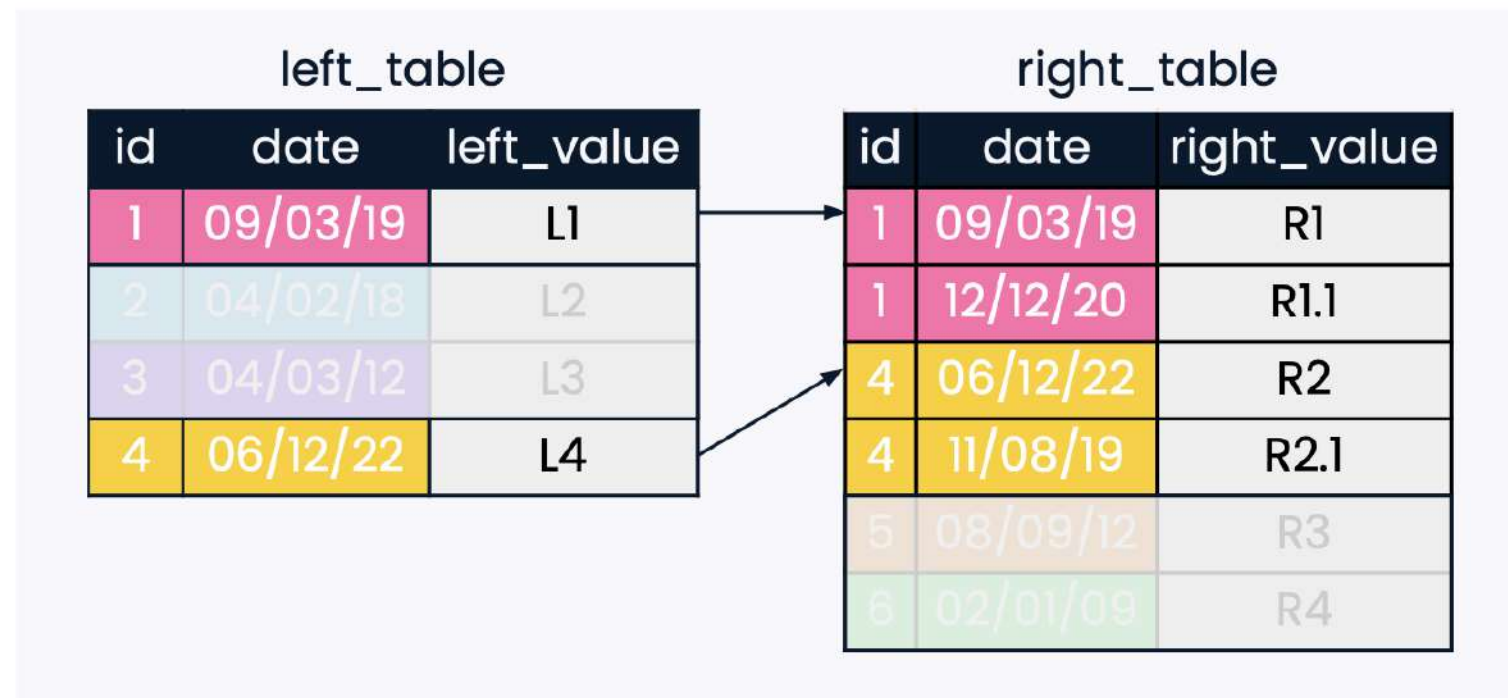
Chaining joins

```
-- SQL query for chaining inner joins
SELECT
    p1.country,
    p1.continent,
    president,
    prime_minister,
    pm_start
FROM prime_ministers as p1
INNER JOIN presidents as p2
    USING(country)
INNER JOIN prime_minister_terms as p3
    USING(prime_minister);
```

What are we joining ON?

```
SELECT *  
FROM left_table  
INNER JOIN right_table  
ON left_table.id = right_table.id;
```

Diagram for an **INNER JOIN** **ON** the **id** field

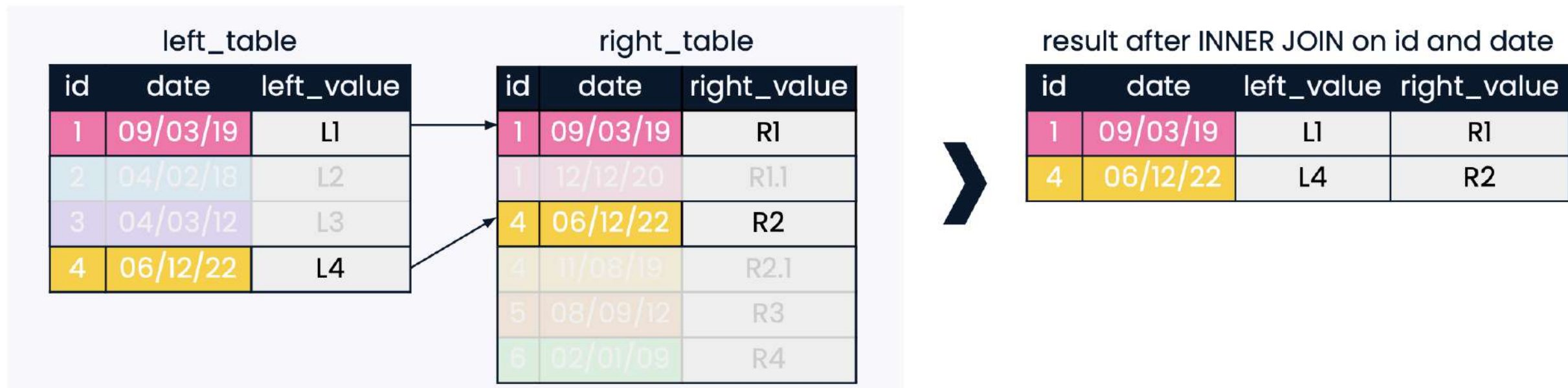


result after INNER JOIN on id			
id	date	left_value	right_value
1	09/03/19	L1	R1
1	12/12/20	L1	R1.1
4	06/12/22	L4	R2
4	11/08/19	L4	R2.1

Joining on multiple keys

```
SELECT *  
FROM left_table  
INNER JOIN right_table  
ON left_table.id = right_table.id  
   AND left_table.date = right_table.date;
```

Diagram for an **INNER JOIN** **ON** the **id** **AND** **date** field



Let's practice!

JOINING DATA IN SQL

LEFT and RIGHT JOINS

JOINING DATA IN SQL

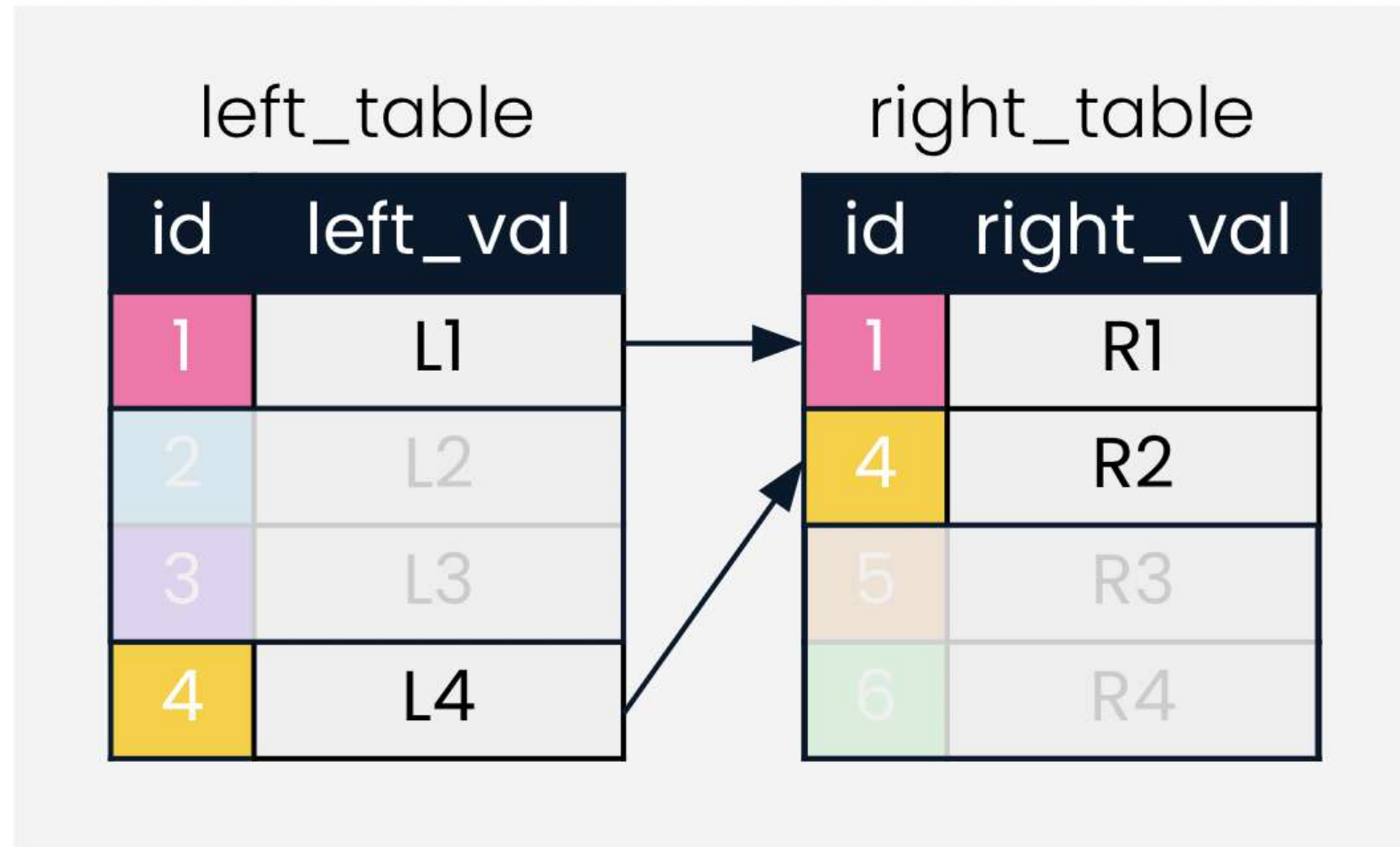


Maham Faisal Khan

Senior Content Developer, Data Science

INNER JOIN diagram

Diagram for an **INNER JOIN** **ON** the **id** field



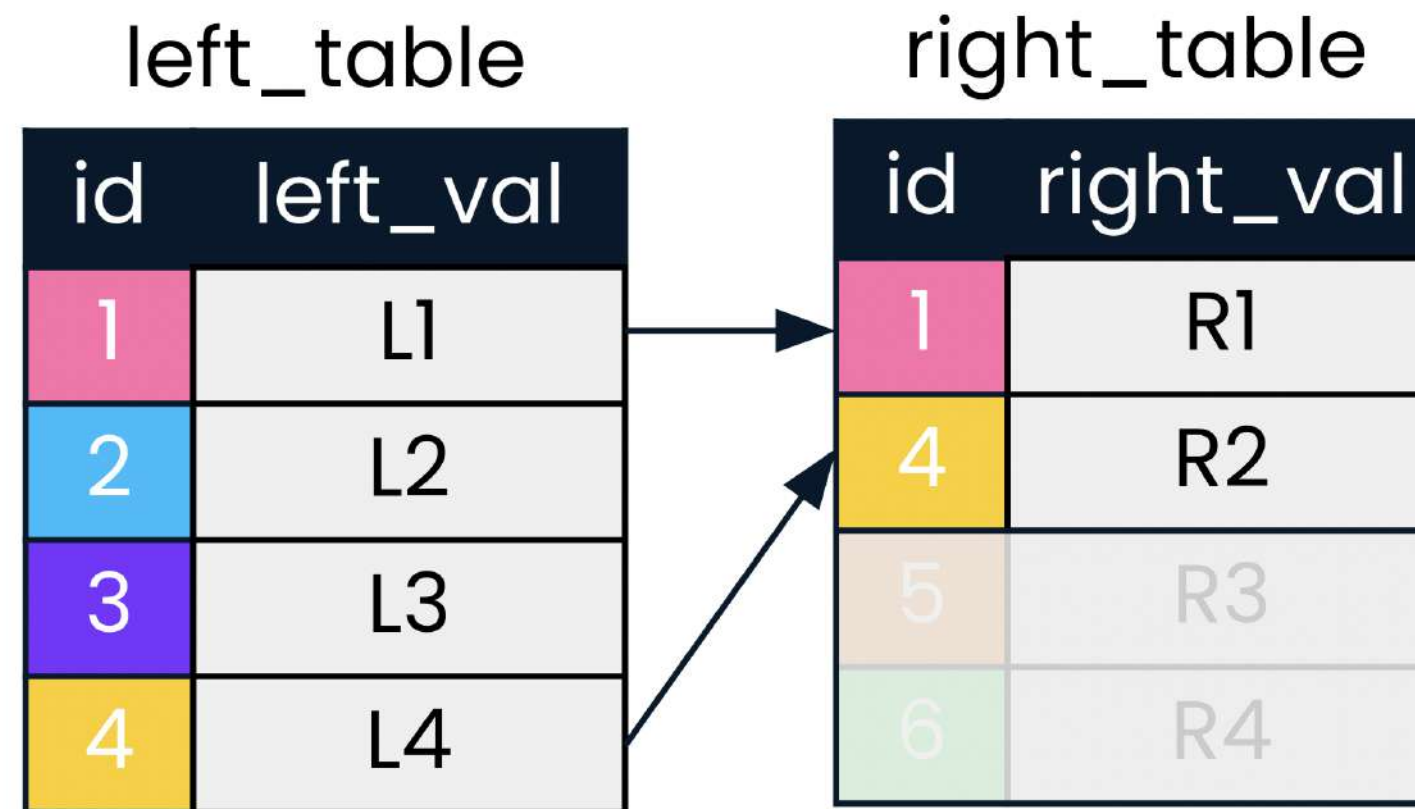
result after INNER JOIN

id	left_val	right_val
1	L1	R1
4	L4	R2

LEFT JOIN initial diagram

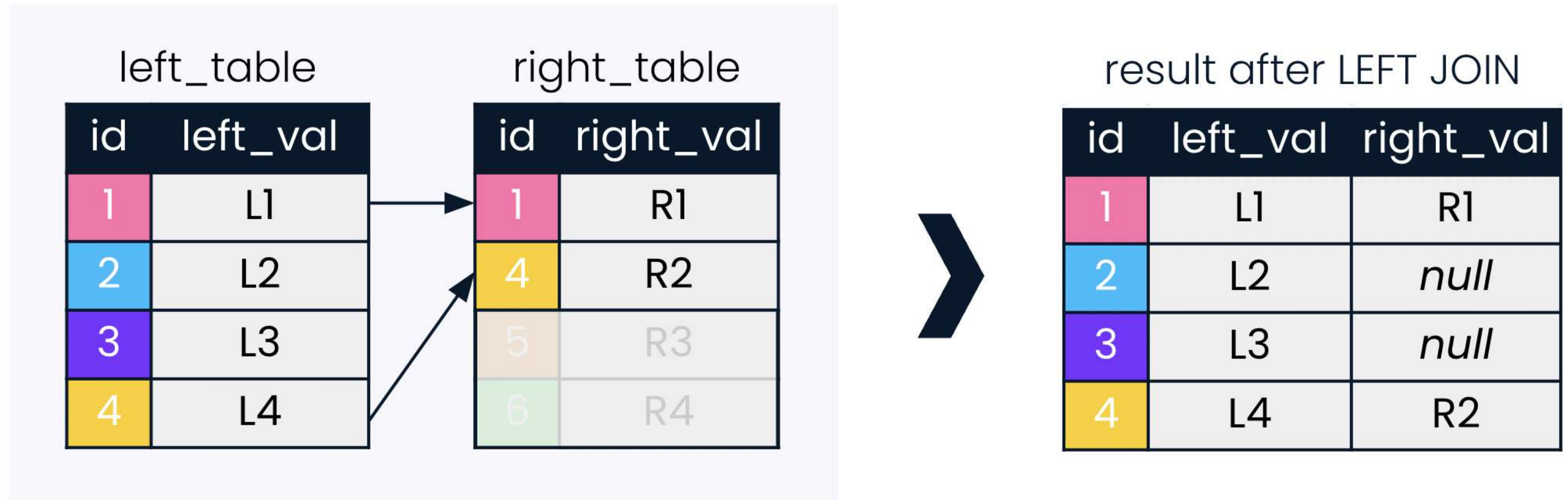
- **LEFT JOIN** will return all records in the left table, and those records in the right table that match on the joining field provided

Diagram for a **LEFT JOIN** **ON** the **id** field



LEFT JOIN diagram

Diagram for a **LEFT JOIN** **ON** the **id** field



LEFT JOIN syntax

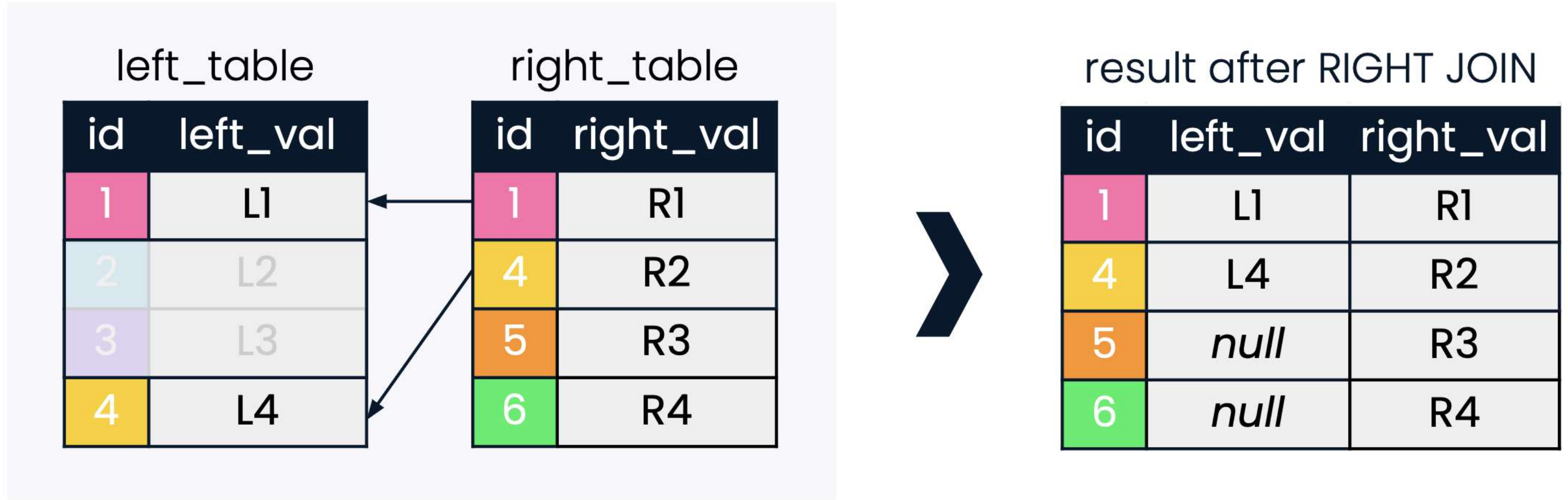
```
SELECT p1.country, prime_minister, president
FROM prime_ministers AS p1
LEFT JOIN presidents AS p2
USING(country);
```

Note. `LEFT JOIN` can also be written as `LEFT OUTER JOIN`.

country	prime_minister	president
Egypt	Mostafa Madbouly	Abdel Fattah el-Sisi
Portugal	Luís Montenegro	Marcelo Rebelo de Sousa
Pakistan	Shehbaz Sharif	Asif Ali Zardari
United Kingdom	Keir Starmer	null
India	Narendra Modi	Droupadi Murmu
Australia	Anthony Albanese	null
...		

RIGHT JOIN

Diagram for a **RIGHT JOIN** **ON** the **id** field



RIGHT JOIN syntax

```
SELECT *  
FROM left_table  
RIGHT JOIN right_table  
ON left_table.id = right_table.id;
```

Note. `RIGHT JOIN` can also be written as `RIGHT OUTER JOIN`.

RIGHT JOIN with presidents and prime ministers

```
SELECT p1.country, prime_minister, president
FROM prime_ministers AS p1
RIGHT JOIN presidents AS p2
USING(country);
```

country	prime_minister	president
Egypt	Mostafa Madbouly	Abdel Fattah el-Sisi
Portugal	Luís Montenegro	Marcelo Rebelo de Sousa
USA	null	Joe Biden
Uruguay	null	Luis Lacalle Pou
Pakistan	Shehbaz Sharif	Asif Ali Zardari
Chile	null	Gabriel Boric
India	Narendra Modi	Droupadi Murmu

LEFT JOIN or RIGHT JOIN?

- `RIGHT JOIN` is less commonly used than `LEFT JOIN`
- Any `RIGHT JOIN` can be re-written as a `LEFT JOIN`
- `LEFT JOIN` feels more intuitive to users when typing from left to right

Let's practice!

JOINING DATA IN SQL

FULL JOINs

JOINING DATA IN SQL



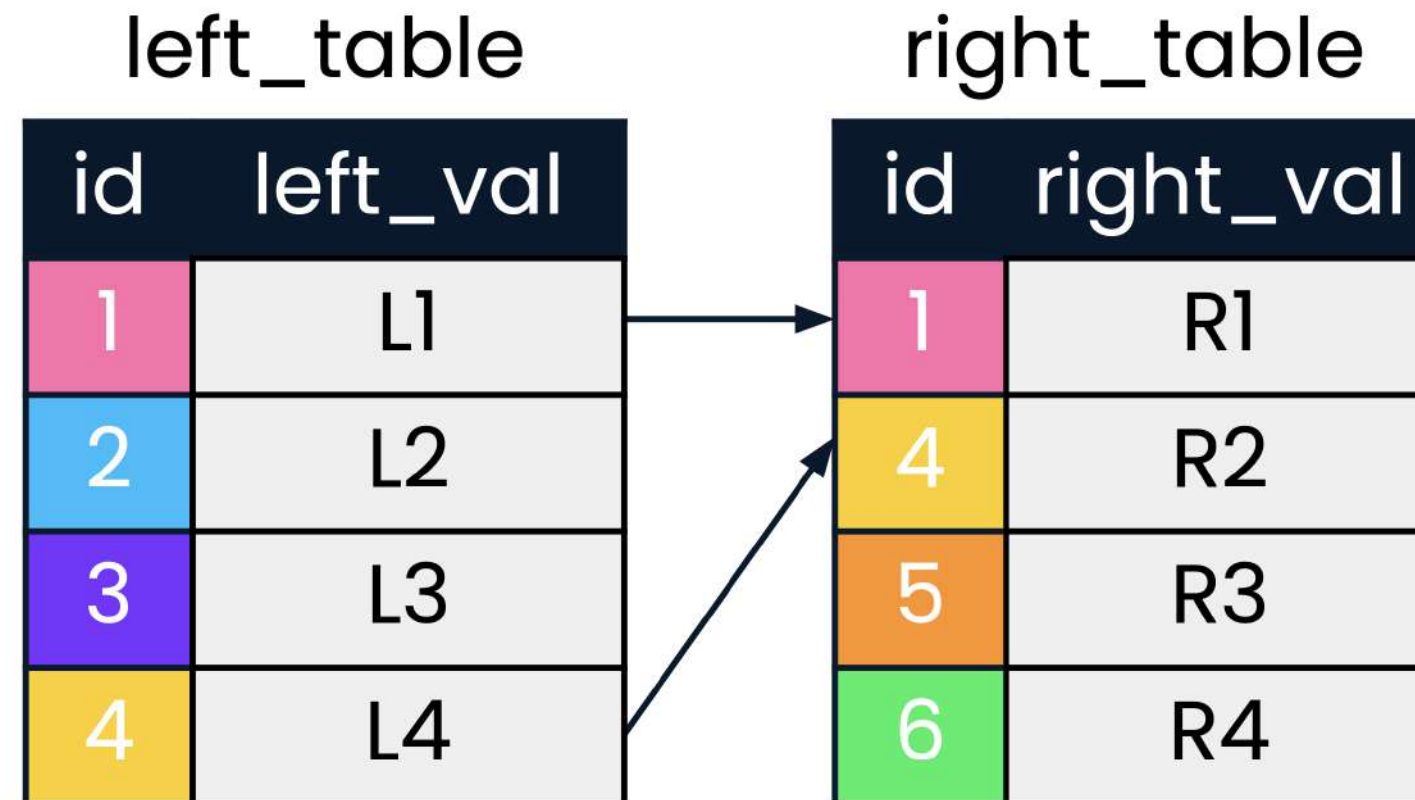
Maham Faisal Khan

Senior Content Developer, DataCamp

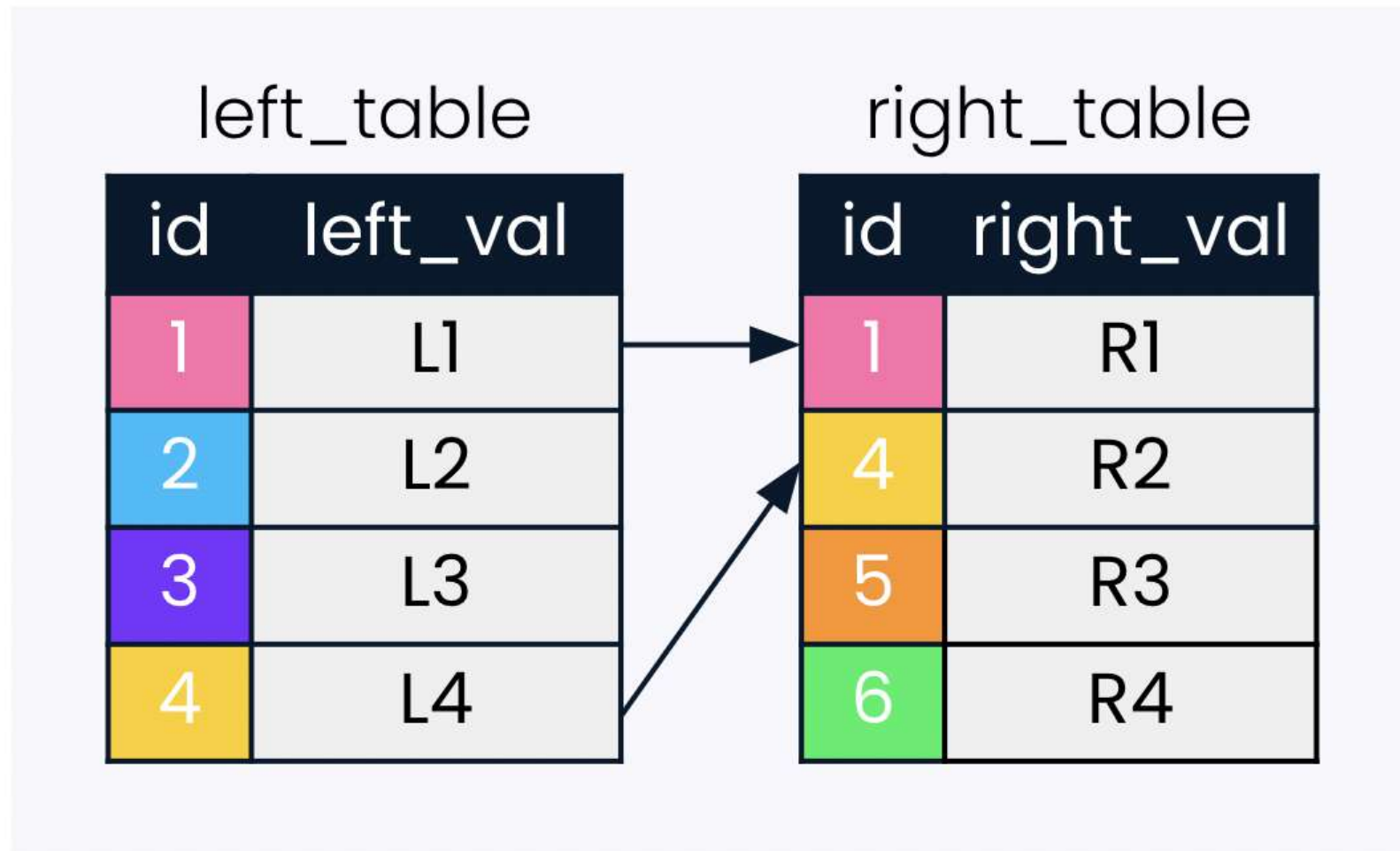
FULL JOIN initial diagram

- A **FULL JOIN** combines a **LEFT JOIN** and a **RIGHT JOIN**

Diagram for a **FULL JOIN ON the id field**



FULL JOIN diagram



result after FULL JOIN

id	left_val	right_val
1	L1	R1
2	L2	null
3	L3	null
4	L4	R2
5	null	R3
6	null	R4

FULL JOIN syntax

```
SELECT left_table.id AS L_id,  
       right_table.id AS R_id,  
       left_table.val AS L_val,  
       right_table.val AS R_val  
FROM left_table  
FULL JOIN right_table  
USING (id);
```

Note. The keyword `FULL OUTER JOIN` can also be used.

FULL JOIN example using leaders database

```
SELECT p1.country AS country, prime_minister, president
```

FULL JOIN example using leaders database

```
SELECT p1.country AS country, prime_minister, president  
FROM prime_ministers AS p1
```

FULL JOIN example using leaders database

```
SELECT p1.country AS country, prime_minister, president  
FROM prime_ministers AS p1  
FULL JOIN presidents AS p2
```

FULL JOIN example using leaders database

```
SELECT p1.country AS country, prime_minister, president
FROM prime_ministers AS p1
FULL JOIN presidents AS p2
ON p1.country = p2.country
LIMIT 10;
```


FULL JOIN example using leaders database

country	prime_minister	president
-----	-----	-----
Egypt	Mostafa Madbouly	Abdel Fattah el-Sisi
Portugal	Luís Montenegro	Marcelo Rebelo de Sousa
Pakistan	Shehbaz Sharif	Asif Ali Zardari
United Kingdom	Keir Starmer	null
USA	null	Joe Biden
Uruguay	null	Luis Lacalle Pou
Chile	null	Gabriel Boric
India	Narendra Modi	Droupadi Murmu
Norway	Jonas Gahr Støre	null
Brunei	Hassanal Bolkiah	null

Let's practice!

JOINING DATA IN SQL

Crossing into CROSS JOIN

JOINING DATA IN SQL

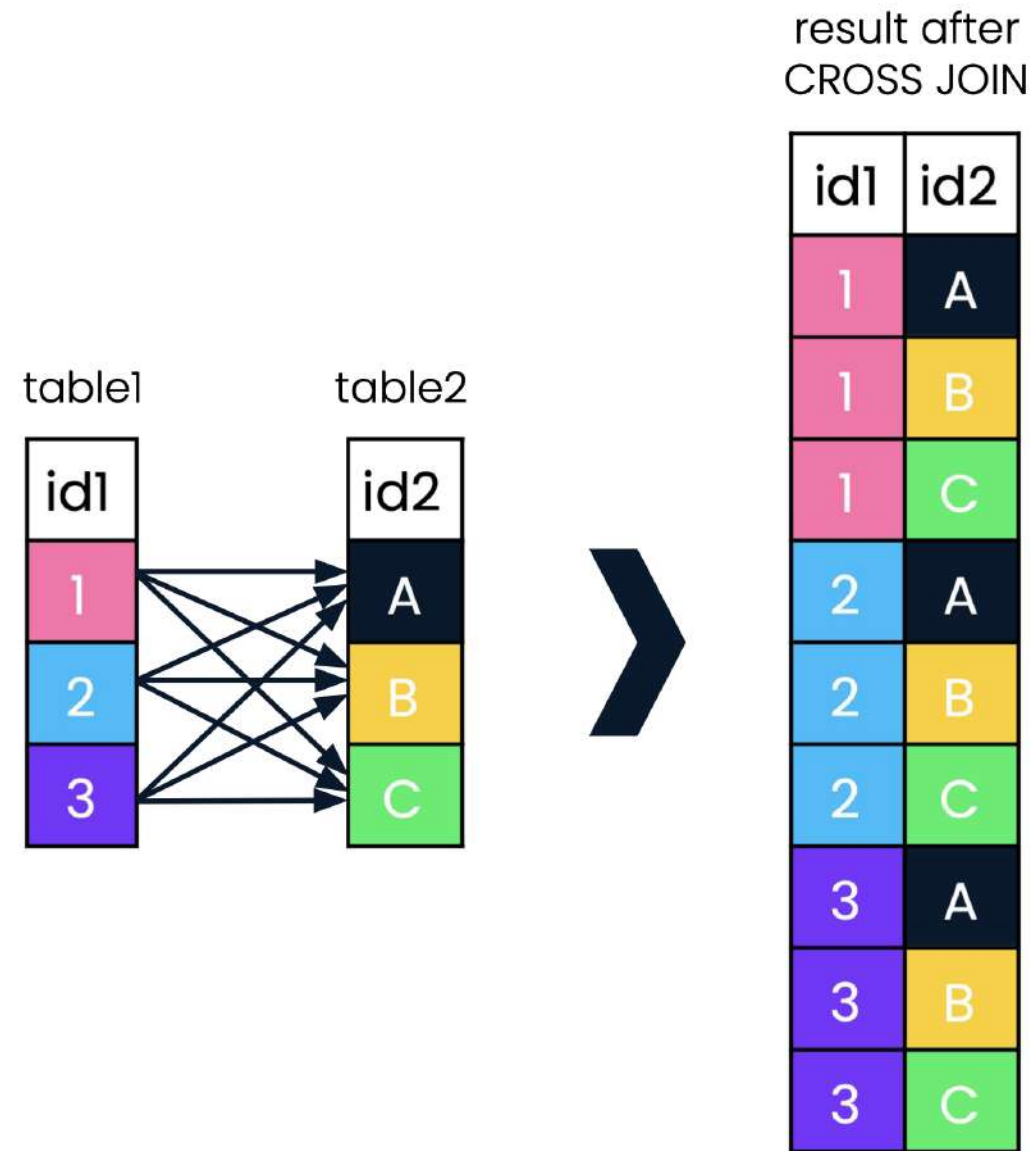


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CROSS JOIN diagram

`CROSS JOIN` creates all possible combinations of two tables.



CROSS JOIN syntax

```
SELECT id1, id2  
FROM table1  
CROSS JOIN table2;
```

Pairing prime ministers with presidents

```
SELECT prime_minister, president
FROM prime_ministers AS p1
CROSS JOIN presidents AS p2
WHERE p1.continent IN ('Asia')
      AND p2.continent IN ('South America');
```

prime_minister	president
Shehbaz Sharif	Luis Lacalle Pou
Narendra Modi	Luis Lacalle Pou
Hassanal Bolkiah	Luis Lacalle Pou
Haitham bin Tarik	Luis Lacalle Pou
Shehbaz Sharif	Gabriel Boric
Narendra Modi	Gabriel Boric
Hassanal Bolkiah	Gabriel Boric
Haitham bin Tarik	Gabriel Boric

Let's practice!

JOINING DATA IN SQL

Self joins

JOINING DATA IN SQL



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Self joins

- Self joins are tables joined with themselves
- They can be used to compare parts of the same table

The `prime_ministers` table from the world leaders database

country	continent	prime_minister
Egypt	Africa	Mostafa Madbouly
Portugal	Europe	Luís Montenegro
Pakistan	Asia	Shehbaz Sharif
United Kingdom	Europe	Keir Starmer
India	Asia	Narendra Modi
Australia	Oceania	Anthony Albanese
Norway	Europe	Jonas Gahr Støre
Brunei	Asia	Hassanal Bolkiah
Oman	Asia	Haitham bin Tarik
New Zealand	Oceania	Christopher Luxon

Prime minister, meet prime minister

```
SELECT
    p1.country AS country1,
    p2.country AS country2,
    p1.continent
FROM prime_ministers AS p1
INNER JOIN prime_ministers AS p2
ON p1.continent = p2.continent
LIMIT 10;
```

Prime minister, meet prime minister

country1	country2	continent
-----	-----	-----
Egypt	Egypt	Africa
Portugal	Spain	Europe
Portugal	Norway	Europe
Portugal	Portugal	Europe
Vietnam	Oman	Asia
Vietnam	Brunei	Asia
Vietnam	India	Asia
Vietnam	Vietnam	Asia
Haiti	Haiti	North America
India	Oman	Asia
India	Brunei	Asia

Prime minister, meet prime minister

```
SELECT
    p1.country AS country1,
    p2.country AS country2,
    p1.continent
FROM prime_ministers AS p1
INNER JOIN prime_ministers AS p2
ON p1.continent = p2.continent
   AND p1.country <> p2.country;
```

The self joined prime ministers table

-----	-----	-----
country1	country2	continent
-----	-----	-----
Portugal	Spain	Europe
Portugal	Norway	Europe
Vietnam	Oman	Asia
Vietnam	Brunei	Asia
Vietnam	India	Asia
India	Oman	Asia
India	Brunei	Asia
India	Vietnam	Asia
Norway	Spain	Europe
Norway	Portugal	Europe
-----	-----	-----

Let's practice!

JOINING DATA IN SQL

Set theory for SQL Joins

JOINING DATA IN SQL

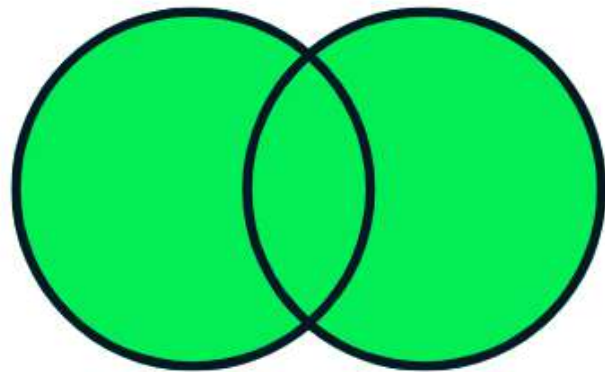


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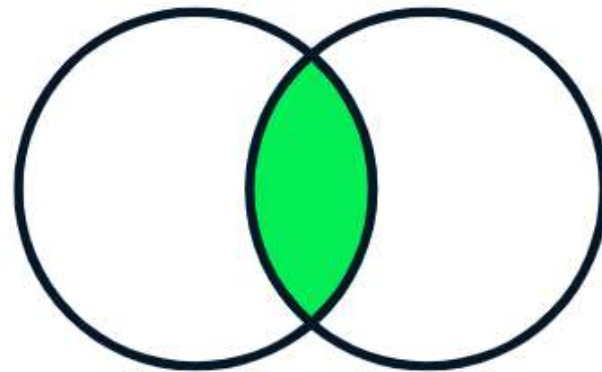
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Venn diagrams and set theory

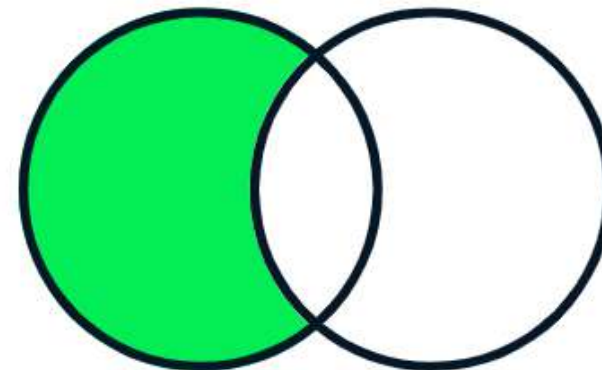
UNION
UNION ALL



INTERSECT

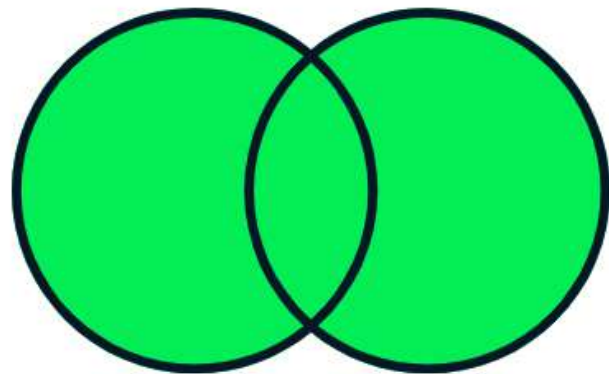


EXCEPT

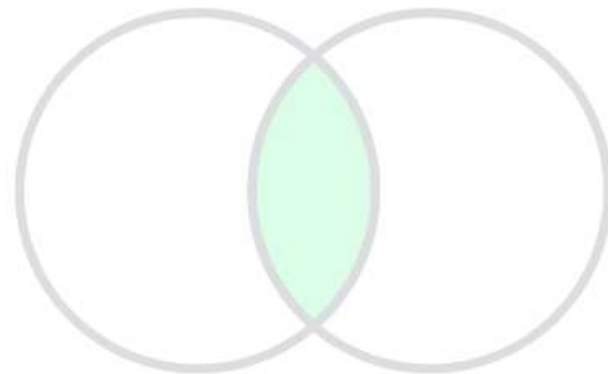


Venn diagrams and set theory

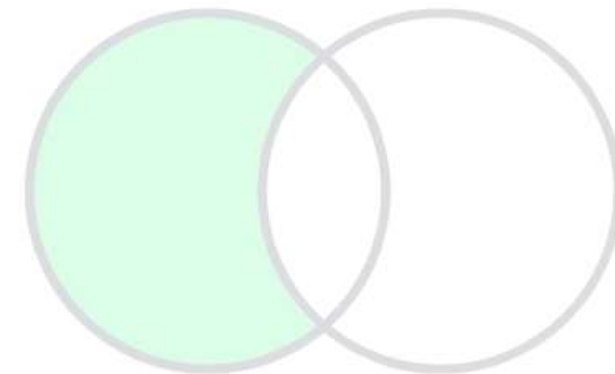
UNION
UNION ALL



INTERSECT

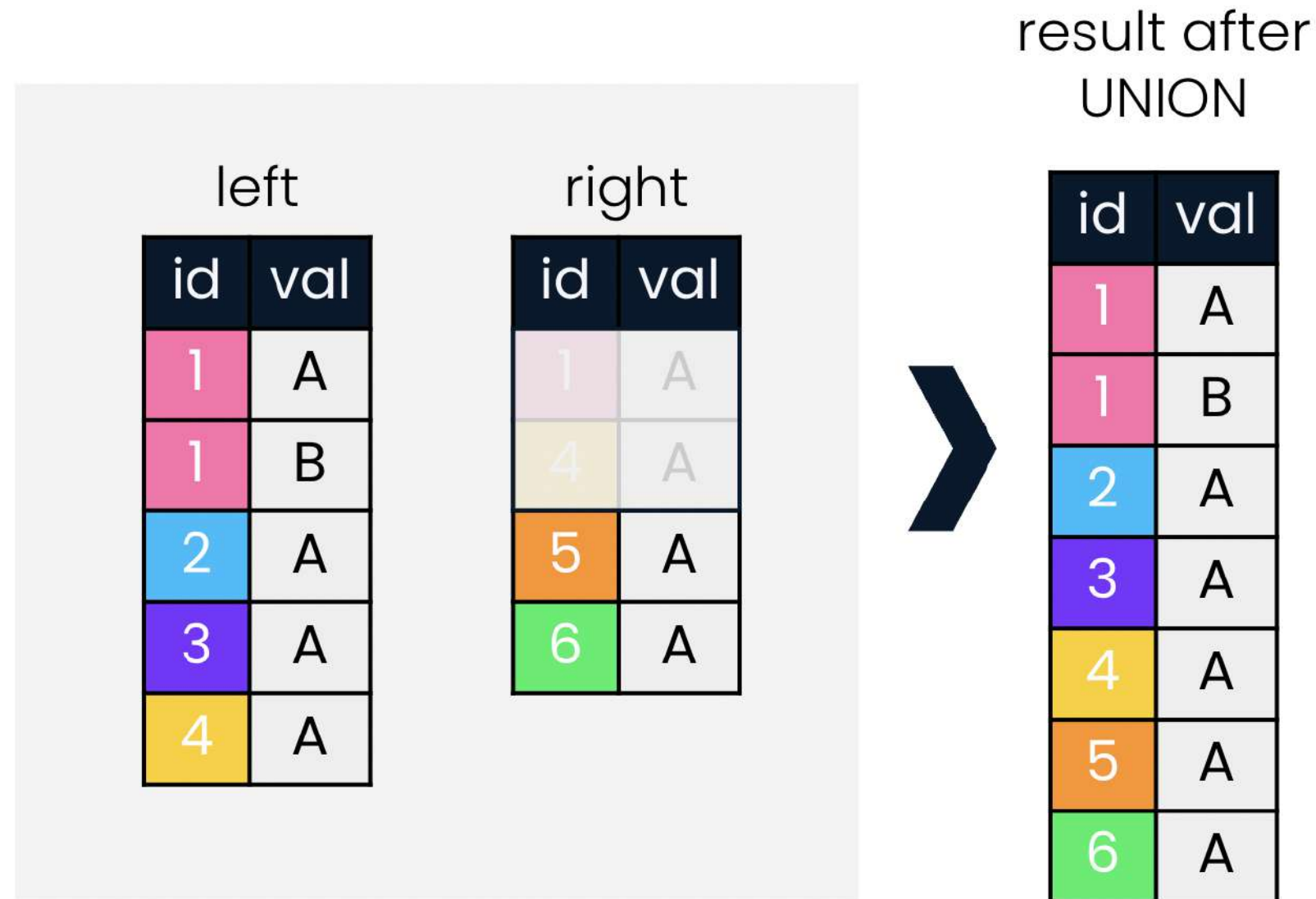


EXCEPT



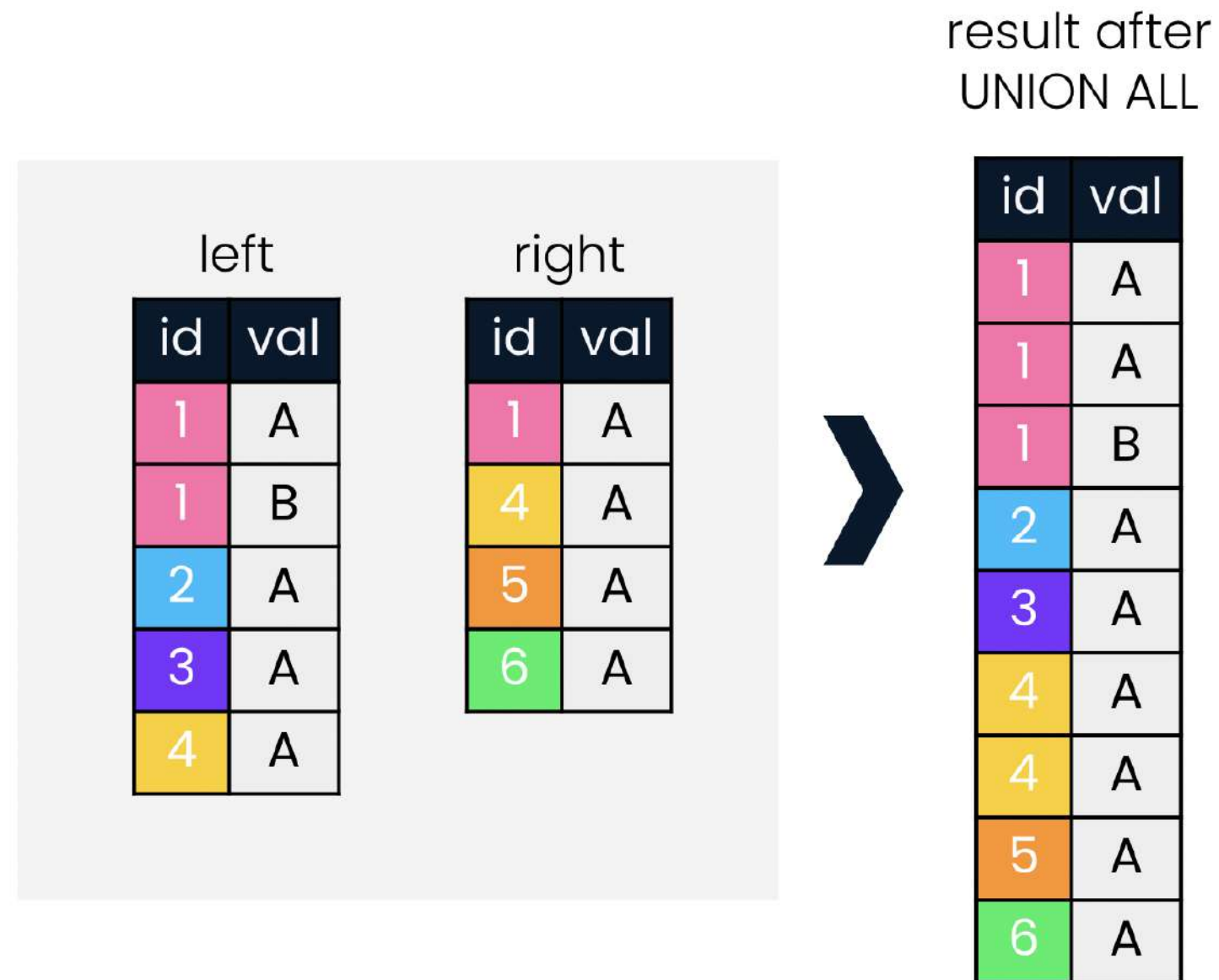
UNION diagram

- **UNION** takes two tables as input, and returns all records from both tables



UNION ALL diagram

- `UNION ALL` takes two tables and returns all records from both tables, including duplicates



UNION and UNION ALL syntax

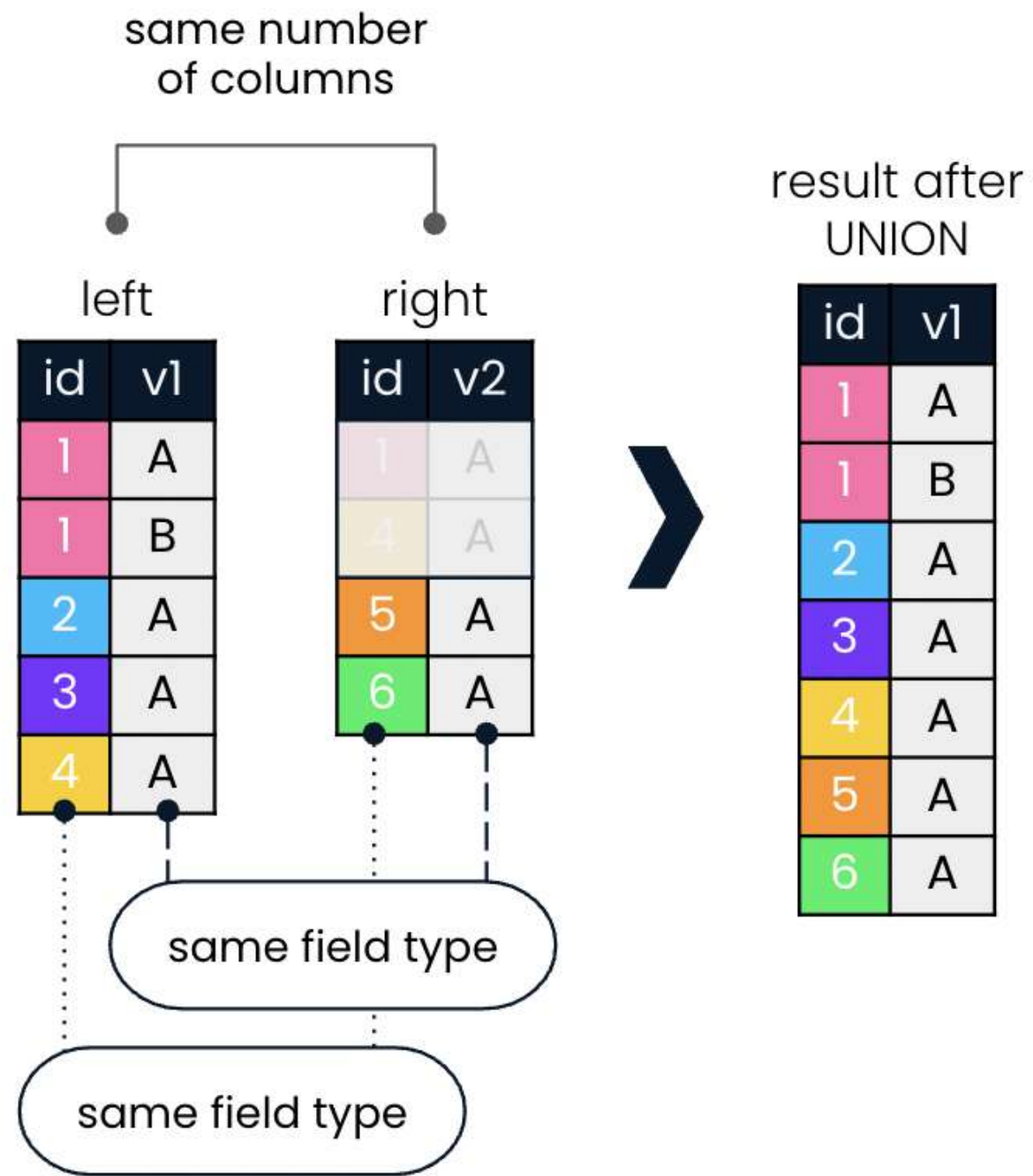
UNION syntax

```
SELECT *  
FROM left_table  
UNION  
SELECT *  
FROM right_table;
```

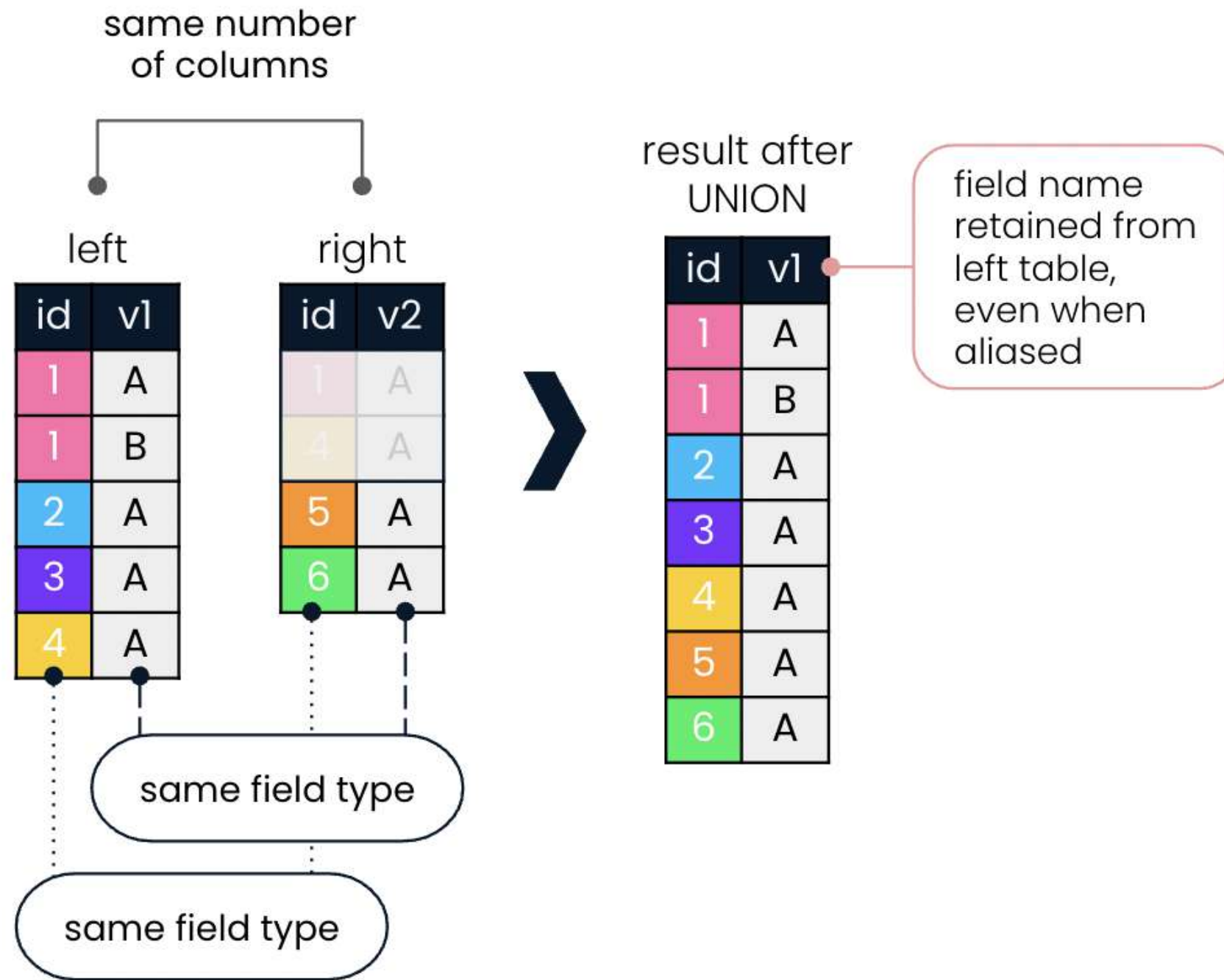
UNION ALL syntax

```
SELECT *  
FROM left_table  
UNION ALL  
SELECT *  
FROM right_table;
```

UNION and UNION ALL syntax



UNION and UNION ALL syntax



To the monarchs table

```
SELECT *  
FROM monarchs;
```

country	continent	monarch
Brunei	Asia	Hassanal Bolkiah
Oman	Asia	Haitham bin Tarik
Norway	Europe	Harald V
United Kingdom	Europe	Charles III

Prime ministers, meet the monarchs

```
SELECT monarch AS leader, country
FROM monarchs
UNION
SELECT prime_minister, country
FROM prime_ministers
ORDER BY country, leader
LIMIT 10;
```


After the UNION

leader	country
Anthony Albanese	Australia
Hassanal Bolkiah	Brunei
Mostafa Madbouly	Egypt
Narendra Modi	India
Christopher Luxon	New Zealand
Harald V	Norway
Jonas Gahr Støre	Norway
Haitham bin Tarik	Oman
Shehbaz Sharif	Pakistan
Luís Montenegro	Portugal

UNION ALL with the leaders

```
SELECT monarch AS leader, country
FROM monarchs
UNION ALL
SELECT prime_minister, country
FROM prime_ministers
ORDER BY leader, country
LIMIT 10;
```

UNION ALL result

leader	country
-----	-----
Anthony Albanese	Australia
Hassanal Bolkiah	Brunei
Hassanal Bolkiah	Brunei
Mostafa Madbouly	Egypt
Narendra Modi	India
Christopher Luxon	New Zealand
Harald V	Norway
Jonas Gahr Støre	Norway
Haitham bin Tarik	Oman
Haitham bin Tarik	Oman

Let's practice!

JOINING DATA IN SQL

At the INTERSECT

JOINING DATA IN SQL

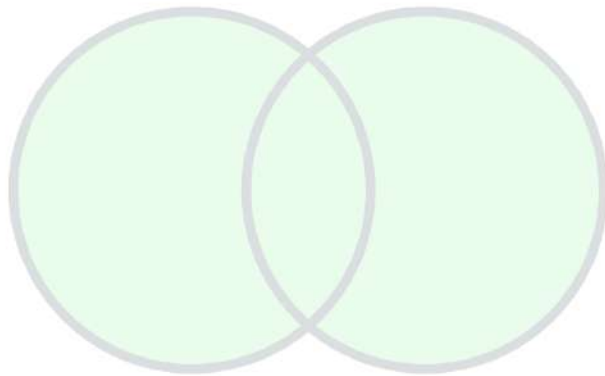


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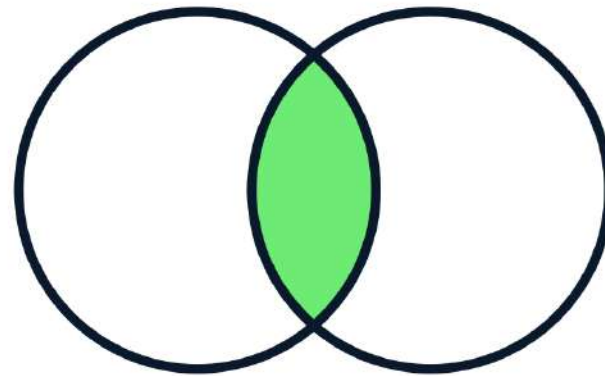
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INTERSECT Venn diagram

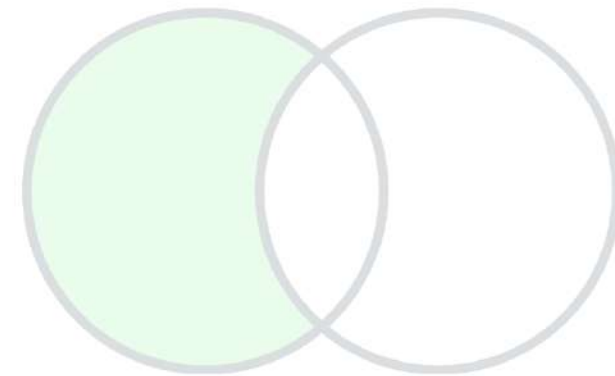
UNION



INTERSECT



EXCEPT



INTERSECT diagram

left_table

id	val
1	N1
1	N1
3	L3
4	L4

right_table

id	val
1	N1
4	R2
5	R3
6	R4



result after
INTERSECT

id	val
1	N1

INTERSECT syntax

```
SELECT id, val  
FROM left_table  
INTERSECT  
SELECT id, val  
FROM right_table;
```

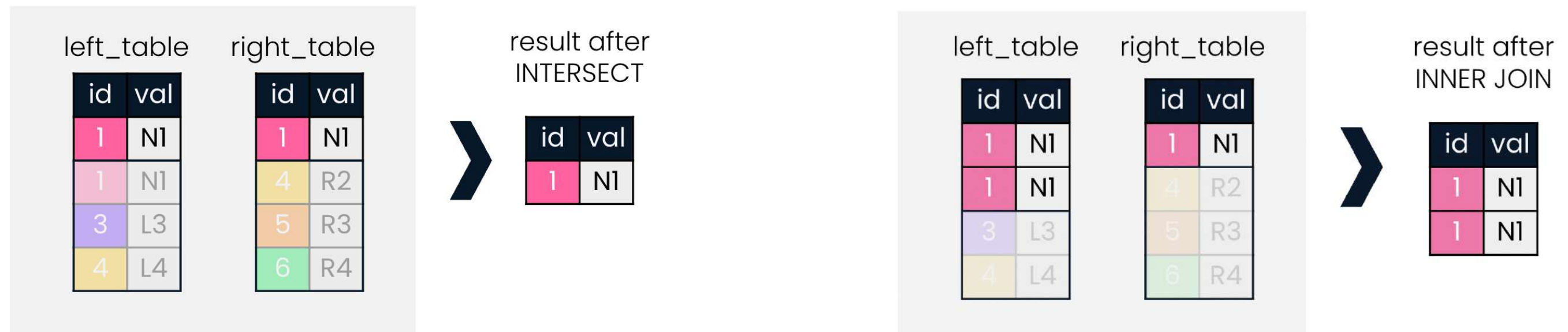

INTERSECT vs. INNER JOIN on two columns

INTERSECT syntax

```
SELECT *  
FROM left_table  
INTERSECT  
SELECT *  
FROM right_table;
```

INNER JOIN syntax

```
SELECT *  
FROM left_table  
INNER JOIN right_table  
ON left.id = right.id  
    AND left.val = right.val
```



Countries with prime ministers and presidents

```
SELECT country as intersect_country
FROM prime_ministers
INTERSECT
SELECT country
FROM presidents;
```

```
| intersect_country |
|-----|
| Egypt            |
| Portugal         |
| Pakistan         |
| India            |
```

INTERSECT on two fields

```
SELECT country, prime_minister AS leader
FROM prime_ministers
INTERSECT
SELECT country, president
FROM presidents;
```

----- -----
country leader
----- -----
----- -----

Countries with prime ministers and monarchs

```
SELECT country, prime_minister AS leader
FROM prime_ministers
INTERSECT
SELECT country, monarch
FROM monarchs;
```

country	leader
Brunei	Hassanal Bolkiah
Oman	Haitham bin Tarik

Let's practice!

JOINING DATA IN SQL

EXCEPT

JOINING DATA IN SQL

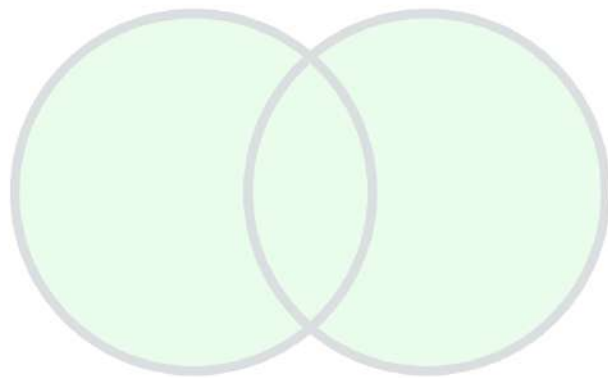


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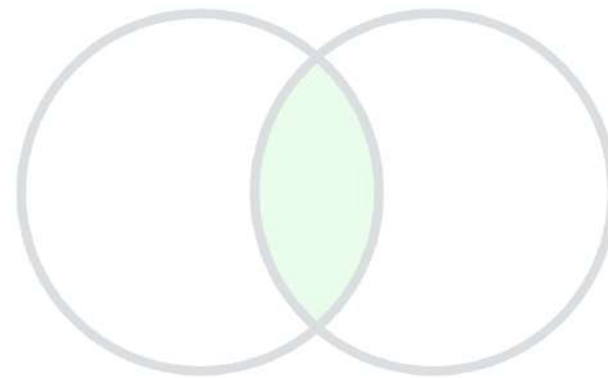
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EXCEPT Venn diagram

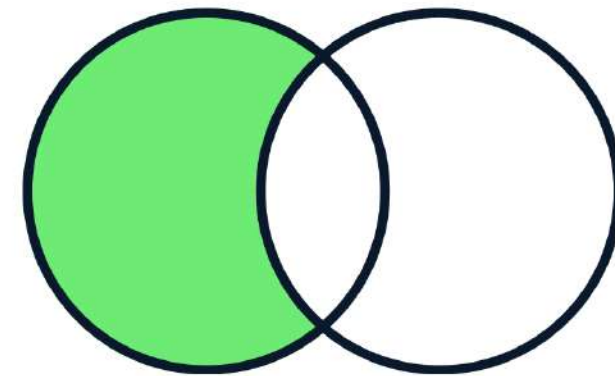
UNION



INTERSECT



EXCEPT



EXCEPT diagram

left_table

id	val
1	N1
1	N1
3	L3
4	L4

right_table

id	val
1	N1
4	R2
5	R3
6	R4



result after
EXCEPT

id	val
3	L3
4	L4

EXCEPT syntax

```
SELECT monarch, country
FROM monarchs
EXCEPT
SELECT prime_minister, country
FROM prime_ministers;
```

----- -----
monarch country
----- -----
Harald V Norway
Charles III United Kingdom
Willem-Alexander Netherlands
----- -----

Let's practice!

JOINING DATA IN SQL

Subquerying with semi joins and anti joins

JOINING DATA IN SQL

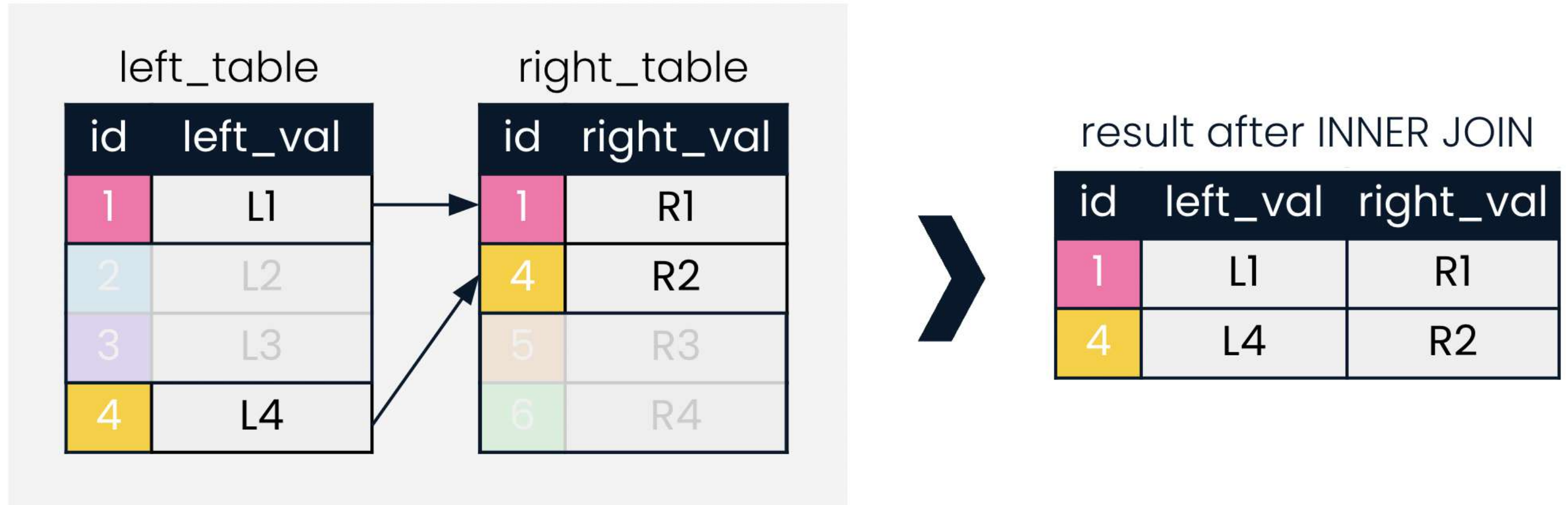
SQL

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Calling all joins

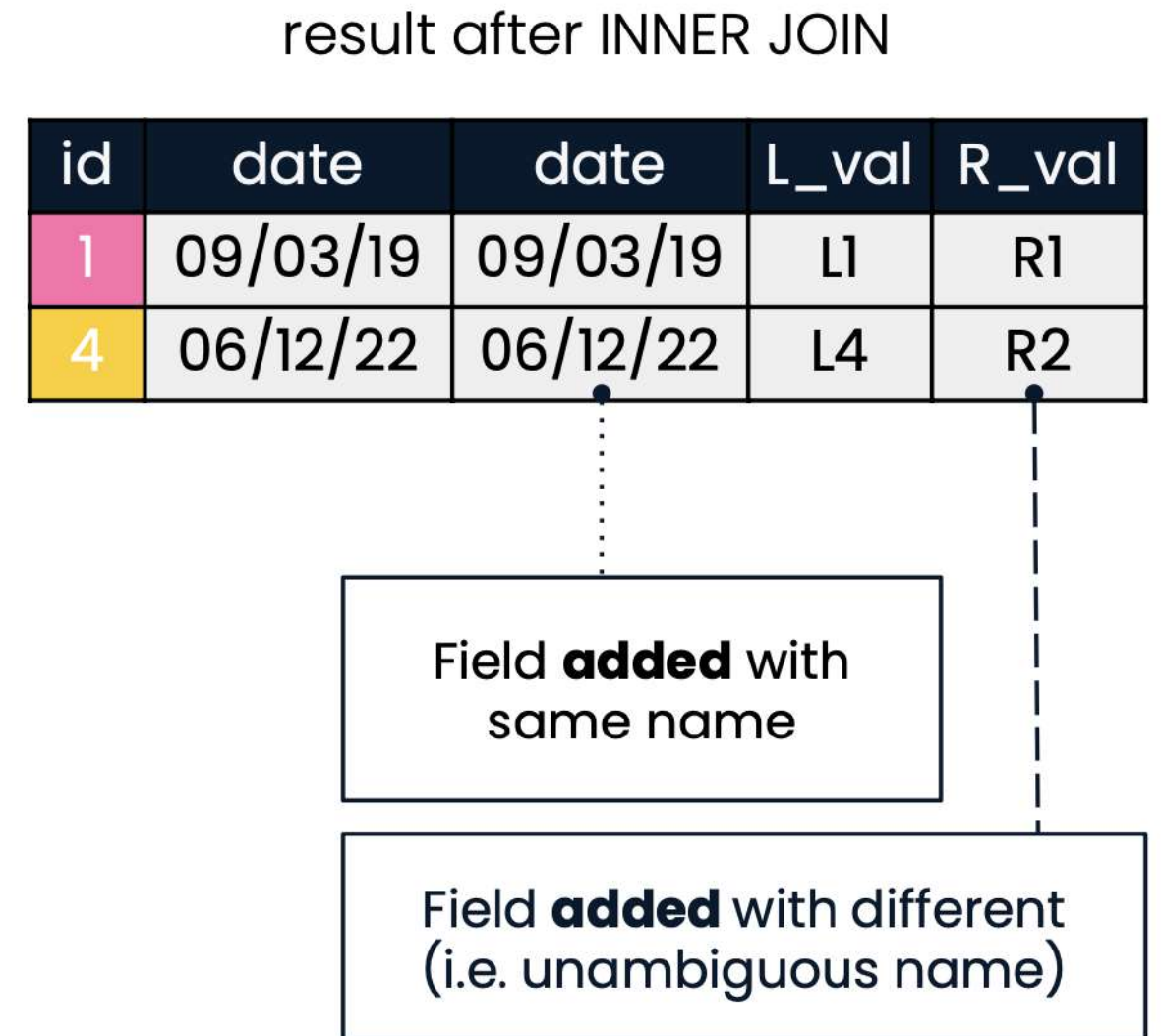
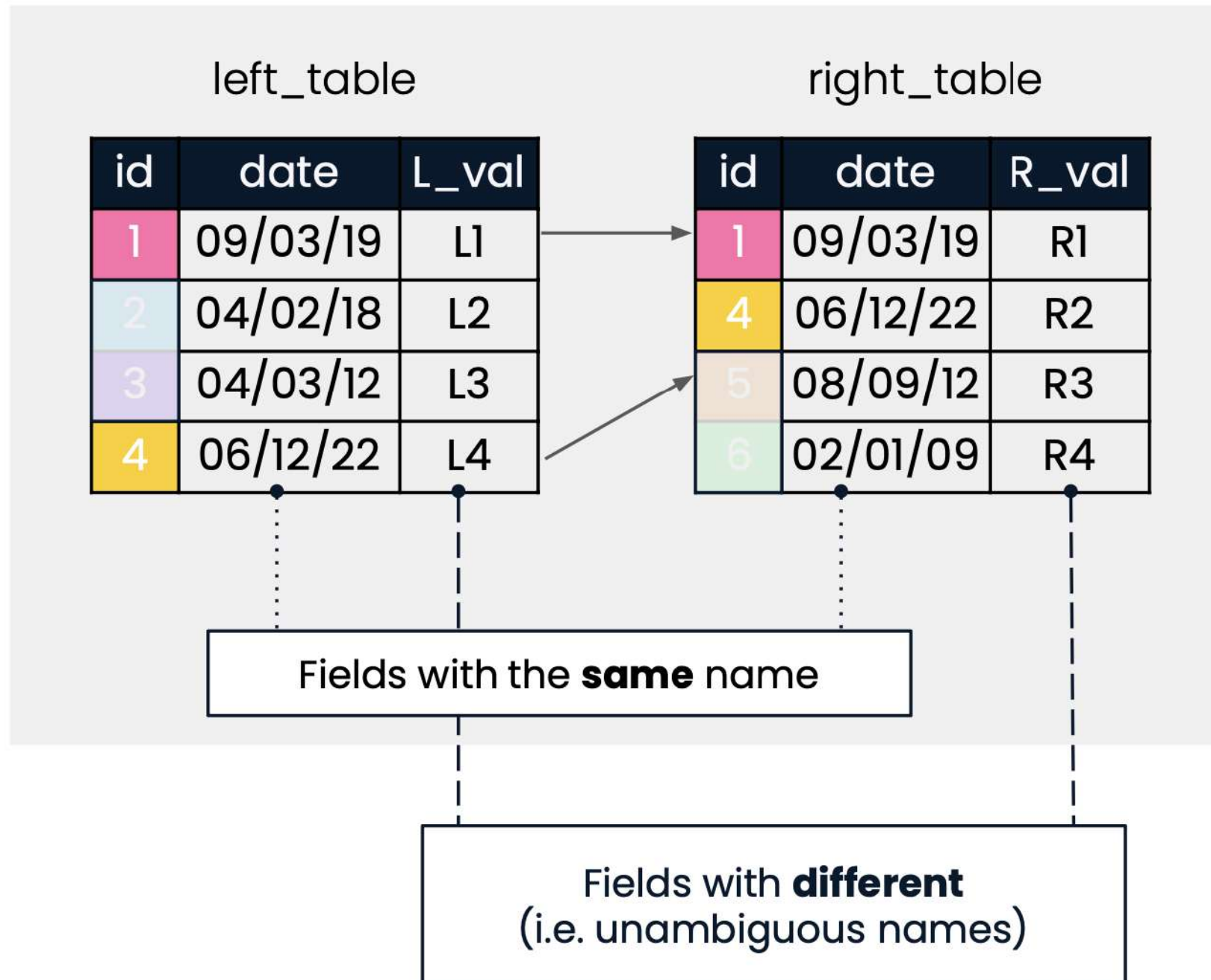
Diagram for an **INNER JOIN** on the **id** field



Additive joins

```
SELECT *  
FROM left_table  
INNER JOIN right_table  
ON left_table.id = right_table.id;
```

Additive joins



Semi join

A **semi join** chooses records in the first table where a condition is met in the second table.

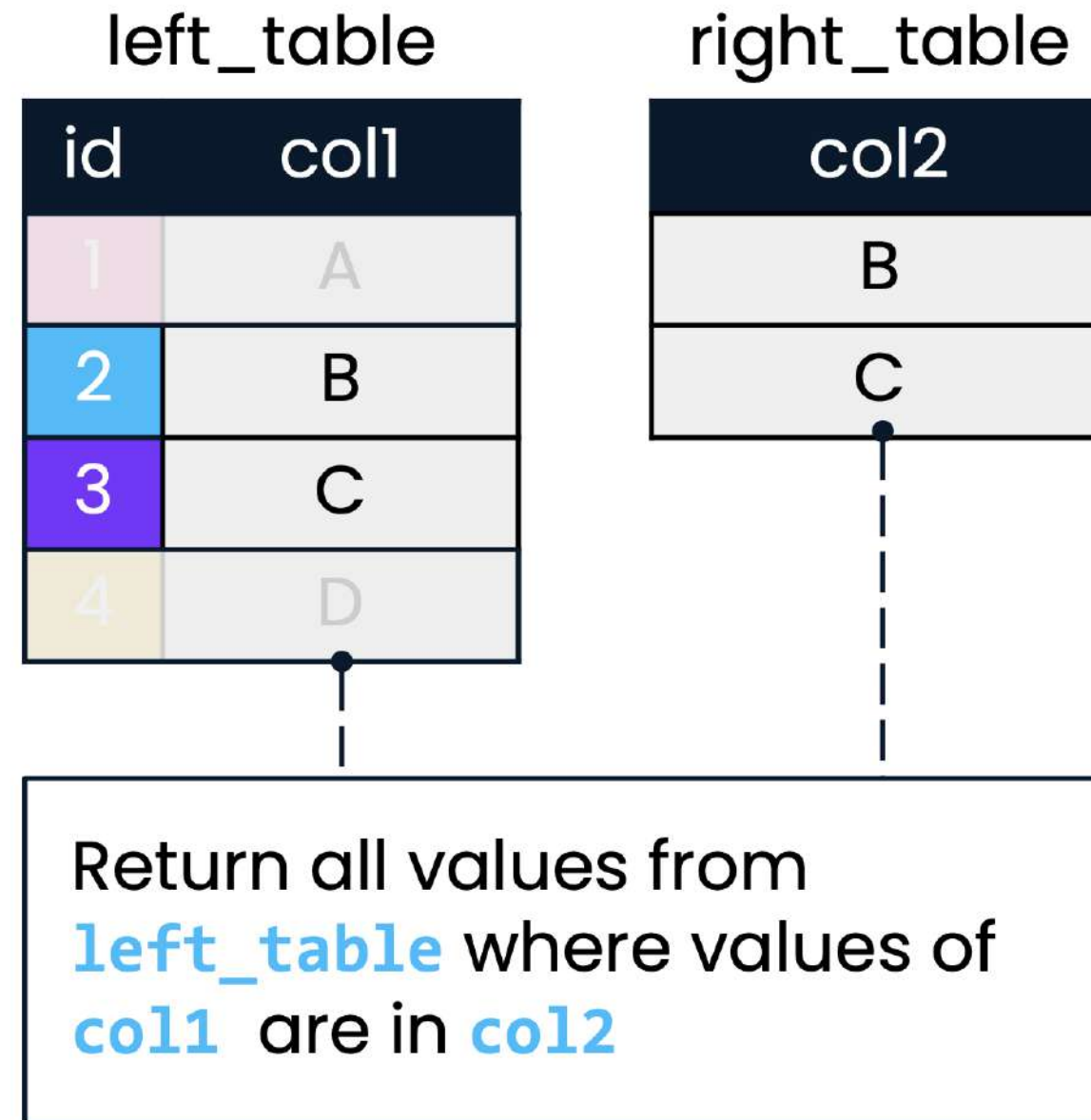
left_table

id	col1
1	A
2	B
3	C
4	D

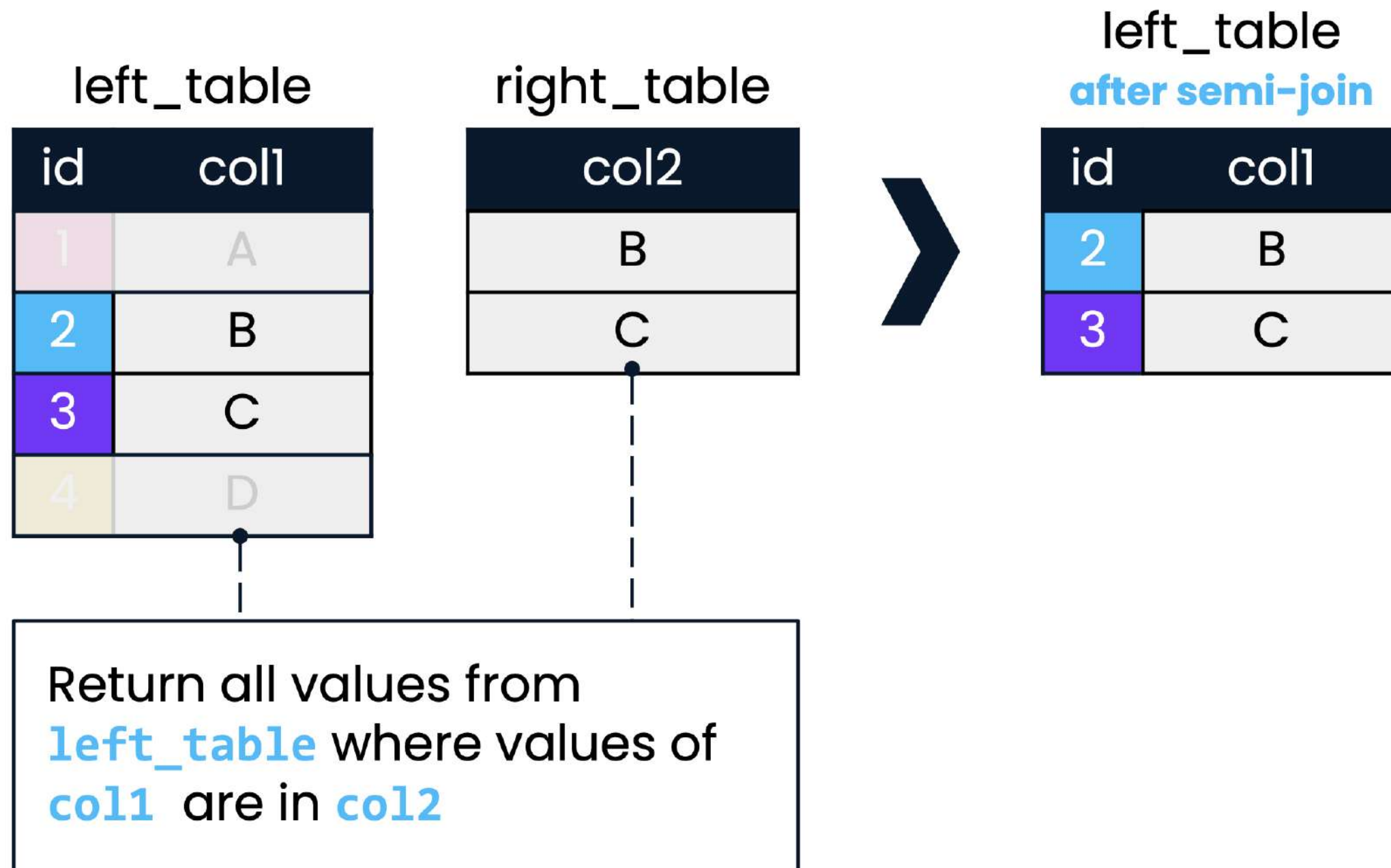
right_table

col2
B
C

Semi join



Semi join



Kicking off our semi join

```
SELECT country, continent, president
FROM presidents;
```

country	continent	president
-----	-----	-----
Egypt	Africa	Abdel Fattah el-Sisi
Portugal	Europe	Marcelo Rebelo de Sousa
USA	North America	Joe Biden
Uruguay	South America	Luis Lacalle Pou
Pakistan	Asia	Asif Ali Zardari
Chile	South America	Gabriel Boric
India	Asia	Droupadi Murmu

Building on our semi join

```
SELECT country
FROM states
WHERE indep_year < 1800;
```

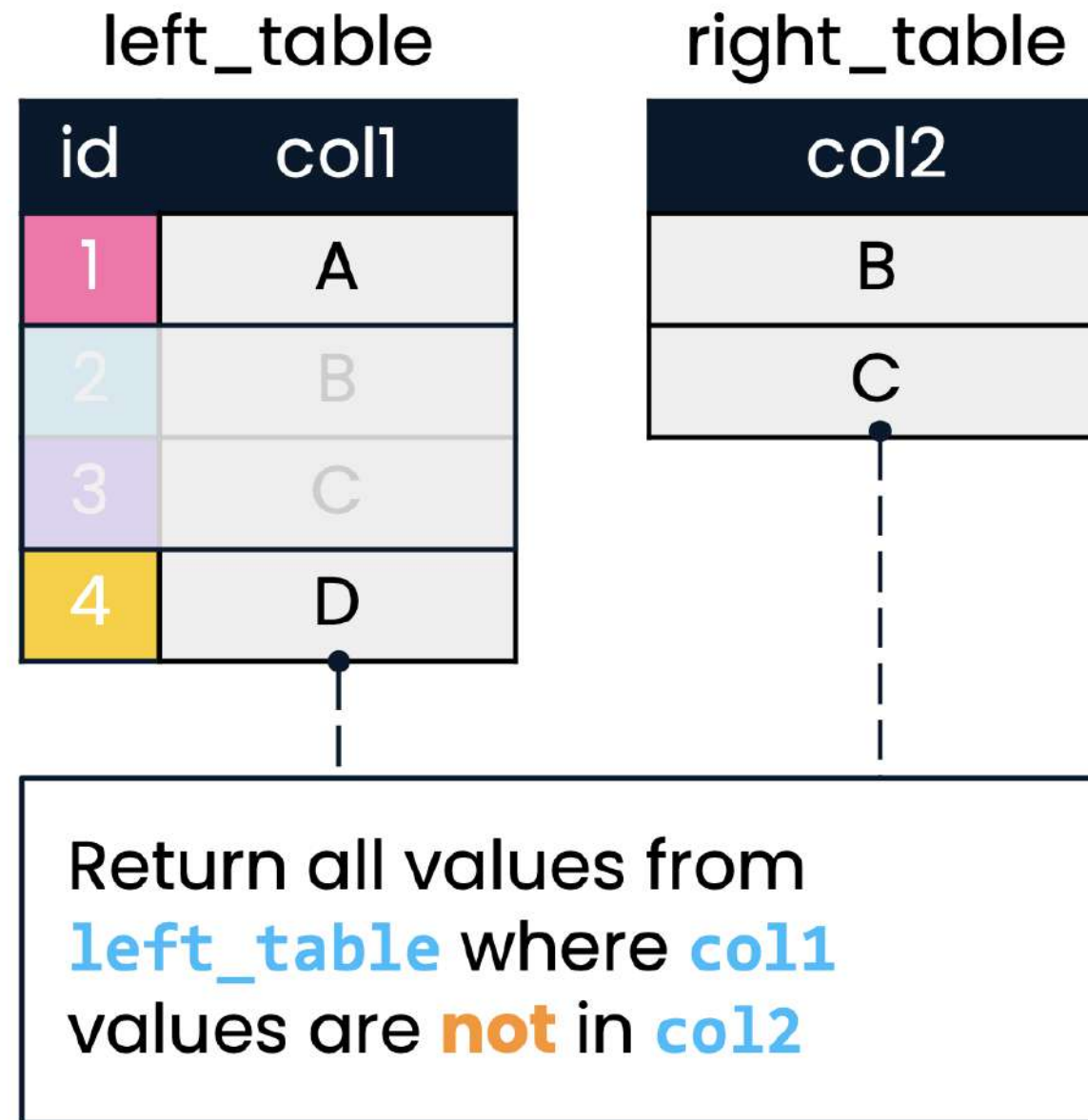
```
|-----|
| country |
|-----|
| Portugal |
| Spain    |
|-----|
```

Finish the semi join (an intro to subqueries)

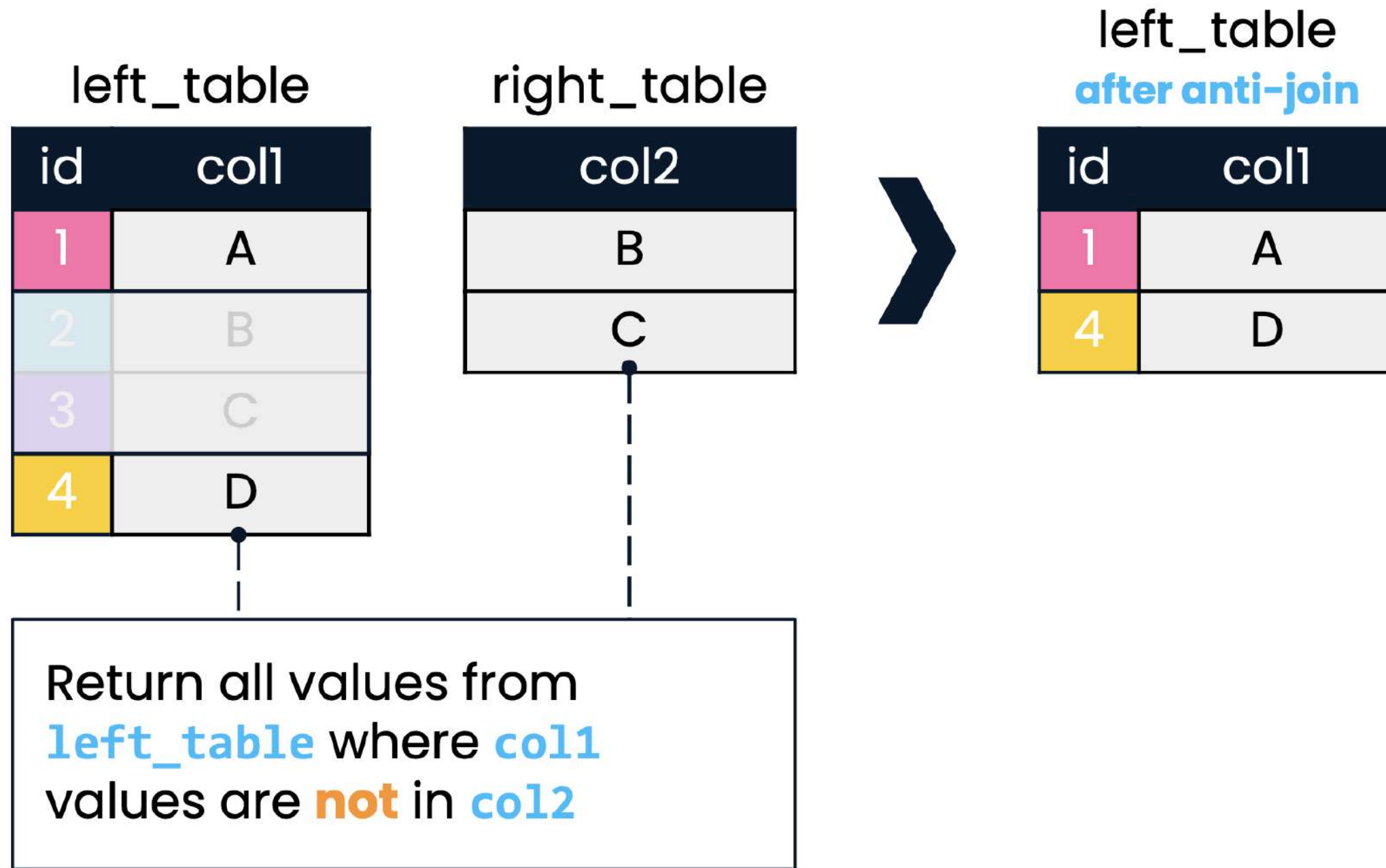
```
SELECT president, country, continent
FROM presidents
WHERE country IN
    (SELECT country
     FROM states
     WHERE indep_year < 1800);
```

-----	-----	-----
president	country	continent
-----	-----	-----
Marcelo Rebelo de Sousa	Portugal	Europe
-----	-----	-----

Anti join



Anti join



An anti join with the presidents

```
SELECT country, president
FROM presidents
WHERE continent LIKE '%America'
      AND country NOT IN
      (SELECT country
       FROM states
       WHERE indep_year < 1800);
```

	country		president	
	-----		-----	
	Uruguay		Luis Lacalle Pou	
	Chile		Gabriel Boric	

Let's practice!

JOINING DATA IN SQL

Subqueries inside WHERE and SELECT

JOINING DATA IN SQL

SQL

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Syntax for subqueries inside WHERE

- All semi joins and anti joins we have seen included a subquery in `WHERE`
- `WHERE` is the most common place for subqueries

Syntax for query using `WHERE IN` statement

```
SELECT *  
FROM some_table  
WHERE some_numeric_field IN (4, 8, 12);
```

Syntax for subqueries inside WHERE

```
SELECT *  
FROM some_table  
WHERE some_field IN  
    (include subquery here);
```

Syntax for subqueries inside WHERE

```
SELECT *  
FROM some_table  
WHERE some_field IN  
    (SELECT some_numeric_field  
     FROM another_table  
     WHERE field2 = some_condition);
```

Subqueries inside SELECT

```
SELECT DISTINCT continent  
FROM states;
```

```
|-----|  
| continent |  
|-----|  
| Africa    |  
| Asia      |  
| Europe    |  
| North America |  
| Oceania   |  
|-----|
```

Subqueries inside SELECT

```
SELECT DISTINCT continent,  
  (SELECT COUNT(*)  
   FROM monarchs  
   WHERE states.continent = monarch.continent) AS monarch_count  
FROM states;
```

continent	monarch_count
Africa	0
Asia	2
Europe	2
North America	0
Oceania	0

Let's practice!

JOINING DATA IN SQL

Subqueries inside FROM

JOINING DATA IN SQL



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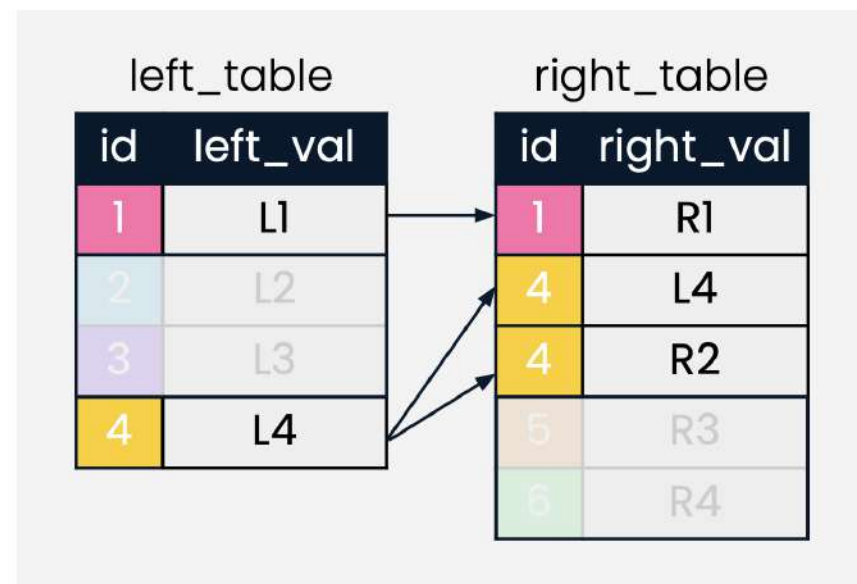
Subqueries inside FROM

```
SELECT continent, MAX(indep_year) AS most_recent
FROM states
GROUP BY continent;
```

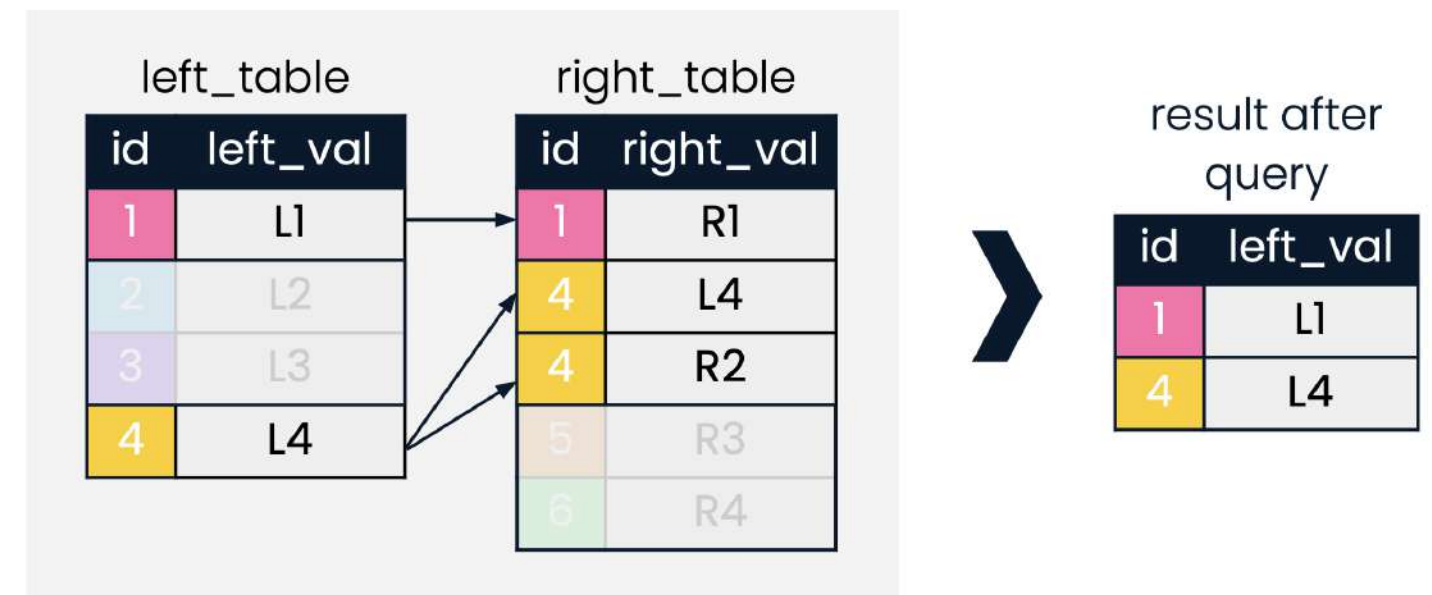
continent	most_recent
-----	-----
Asia	1984
Europe	1814
Oceania	1901
North America	1776
South America	1818

Focusing on records inside monarchs

```
SELECT left_table.id, left_val  
FROM left_table, right_table  
WHERE left_table.id = right_table.id
```



```
SELECT DISTINCT left_table.id, left_val  
FROM left_table, right_table  
WHERE left_table.id = right_table.id
```



Finishing off the subquery

```
-- Query to return continents with monarchs and the year the most recent country gained independence
SELECT DISTINCT monarchs.continent, sub.most_recent
FROM monarchs,
    (SELECT
        continent,
        MAX(indep_year) AS most_recent
    FROM states
    GROUP BY continent) AS sub
WHERE monarchs.continent = sub.continent
ORDER BY continent;
```

continent	most_recent	
-----	-----	
Asia	1984	
Europe	1814	

Let's practice!

JOINING DATA IN SQL

The finish line

JOINING DATA IN SQL



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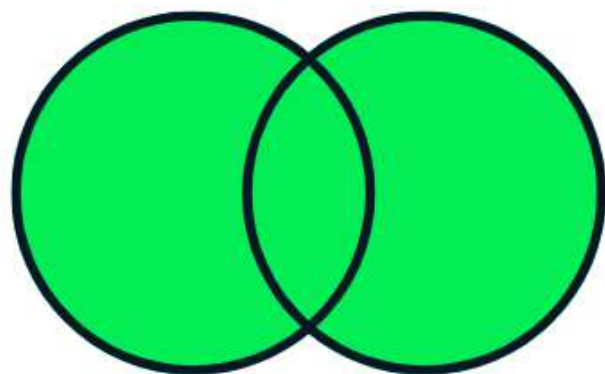
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Types of joins

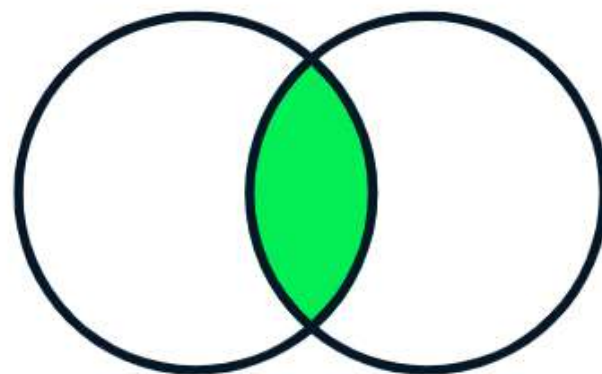
1. `INNER JOIN` , or just `JOIN`
2. Outer join
 - `LEFT JOIN`
 - `RIGHT JOIN`
 - `FULL JOIN`
3. `CROSS JOIN`
4. Semi join / anti join
5. Self join

Set operations

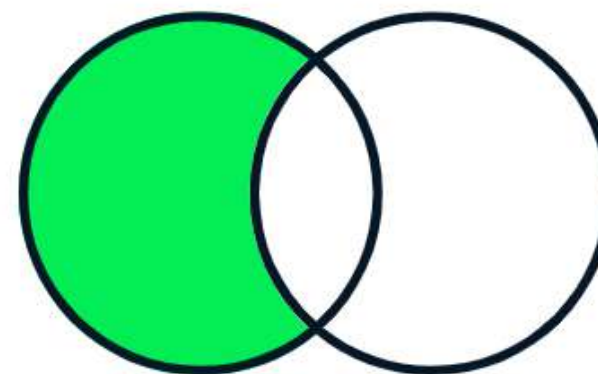
UNION
UNION ALL



INTERSECT



EXCEPT



Types of basic subqueries

- Subqueries inside `SELECT` clauses
- Subqueries inside `WHERE` clauses
- Subqueries inside `FROM` clauses

WHERE to from here?

- DataCamp's Intermediate SQL course
- Projects, practice exercises and competitions
- Workspace

The finish line!

JOINING DATA IN SQL